DYSANOS II

Smooth SANOS-Type Equity Option Market Models with Static and Dynamic No-Arbitrage Guarantees

Density-coordinate drift restrictions, Black product lifts, front settlement and consistent recalibration

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Abstract

A useful equity option market simulator must reprice the same surviving contracts as the market evolves, generate realistic statistical scenarios, and avoid both cross-sectional and dynamic arbitrage. We develop a constructive route from a finite discrete-local-volatility (DLV) market to smooth SANOS-type option dynamics. First, we show that the Wiese density-coordinate drift restriction is the exact non-front-maturity reformulation of Wissel's finite-option drift restriction. Second, we identify the two boundary questions that this identity does not settle: terminal consistency at the front expiry and prices between quoted strikes. We construct a bounded latent martingale X and an independent Black martingale Y, and price calls on $S = XY$ by conditional expectation. At model expiries this gives finite Black mixtures, smooth in strike, with true-martingale prices and exact settlement. An explicit finite-state construction supplies a finite-mixture surface for every maturity; a Brownian revelation construction supplies continuous spot paths at prescribed model expiries. The product theorem also extends to independent stochastic-volatility smoothing, including a concrete model with stochastic volatility of volatility. The Black kernel is then replaced by its full conditional pricing kernel. The induced smooth drift restriction has the Carmona–Nadtochiy covariance form, with a predictable jump correction when required.

Exact agreement with protected DLV quotes is obtained by joint calibration, not by adding positive smoothing to unchanged atomic weights. We prove local feasibility for strictly admissible finite books and distinguish initial quote matching from preservation of a preassigned entire grid process. Finally, we formulate a CRC-inspired historical calibration programme: construct admissible Q-dynamics first, estimate physical dynamics only through permitted risk premia, and preserve surviving prices at parameter resets. Structural identities, numerical discretisation and sample-based OCE diagnostics are treated separately. A dedicated Bates analysis proves the finite-Black-mixture representation obstruction, derives an exact continuous mixing representation, and distinguishes these oracle-specific facts from the existence of finite SANOS dynamics. It also supplies exact projection diagnostics, terminal rigidity results and alternative dynamic completions of directly calibrated marginals.

Keywords. Option market simulation; SANOS; discrete local volatility; martingale measures; smooth option surfaces; consistent recalibration; deep hedging.

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1 Introduction

1.1 From equity option simulation to smooth dynamic surfaces

Deep hedging learns trading rules from repeated market scenarios rather than from a prescribed analytic hedge. If options are themselves hedging instruments, the training environment must generate their joint evolution with spot, including the returns of each *fixed cash-strike, fixed calendar-expiry* contract held by the strategy. Historical observations alone give relatively few independent episodes, particularly in stressed regimes. A market simulator therefore becomes part of the economic specification of the learning problem: defects in simulated prices can be learned as artificial trading opportunities.

Wiese, Bai, Wood and Buehler's *Deep Hedging: Learning to Simulate Equity Option Markets* [1] established an important link between data-driven scenario generation and the DLV representation. Positivity in DLV coordinates protects the discrete cross-section against static arbitrage. The learned statistical transition is a separate object, however; a statically admissible decoder alone does not construct one equivalent measure under which every surviving option is a martingale. This distinction is particularly important when the quoted lattice rolls with spot and time.

There is also an interpolation problem. For $K = (1-\theta)K_i + \theta K_{i+1}$, convexity requires

$$C(T,K) \le (1-\theta)C(T,K_i) + \theta C(T,K_{i+1}), \qquad 0 \le \theta \le 1. \tag{1.1}$$

Thus the linear chord is the *most expensive* convex interpolation between its endpoints. Here "expensive" means a high option price, not high computational complexity. Linear interpolation is cheap to compute. Its economic drawback is the upper-envelope price between quotes and the concentration of the implied distribution at grid knots; it does not produce a smooth density. A hedging agent trained on this envelope may face systematically distorted off-grid trading costs. This concern connects to Buehler's *Expensive martingales* [3].

SANOS [4] replaces this envelope by positive mixtures of smooth pricing kernels, with a particularly transparent Black implementation. Its combination of smoothness, inexpensive evaluation and constrained calibration makes it a natural decoder for an equity option market simulator. At a common set of fitted quotes a convex smooth fit lies below the corresponding linear chords, and can therefore supply less expensive off-grid prices. This is a comparison after calibration; smoothing a *fixed* atomic distribution instead increases convex-payoff values. These two statements concern different operations.

Buehler, Horváth and Kratsios's DYSANOS [5], referred to here as *DYSANOS I*, takes the first generative step using smooth SANOS surfaces. It combines a static arbitrage-free representation with estimated statistical state dynamics and investigates dynamic arbitrage empirically. The present paper develops *DYSANOS II*: a smooth SANOS-type market model in which common risk-neutral dynamics and settlement are part of the construction. Historical realism remains an estimation objective, not a consequence of the no-arbitrage theorem.

1.2 Related literature and the precise gap

The market-model approach specifies dynamics of traded options alongside spot. Wissel's 2007 finite-option paper [6], and the corresponding work collected in his thesis [7], are the relevant discrete predecessors. They should be distinguished from Schweizer–Wissel's continuous-strike, single-maturity model [8] and their fixed-strike term-structure model [9]. Section 2 identifies an exact change of coordinates for the non-front finite-grid drift restriction. It does not identify all front conventions or all admissible coefficient classes across these papers.

Carmona–Nadtochiy [11, 12] develop a local-volatility codebook with a drift restriction and a spot-consistency boundary. Their subsequent tangent Lévy models [13], and the option HJM formulation of Kallsen–Krühner [14], treat jump codebooks. They make clear why positive

cross-sectional parameters are insufficient: the evolving codebook must remain a conditional description of the same market. The density-product identity derived below is the finite-grid and continuous-strike expression of this compatibility.

Gambara–Teichmann's consistent recalibration models [15] permit parameters of a tractable risk-neutral model to change while a Hull–White-type extension compensates so that the evolved option surface is retained. Their generalised Bates example supplies inspiration for our *order of operations*: evolve, propose a parameter update, then solve a price-preserving correction. Our correction acts on conditional distributions and SANOS coordinates. It is not obtained by substituting SANOS parameters into their theorem without checking admissibility.

Cont–da Fonseca [16] and Cont–Fonseca–Durrleman [17] establish an empirical factor perspective on implied-volatility dynamics. Cont–Vuletić [18] and VolGAN [19] pursue realistic scenario generation with cross-sectional arbitrage control. Their contribution should not be described as ignoring arbitrage or surface dynamics. The distinction relevant here is that cross-sectional control and statistical fidelity do not, on their own, furnish the conditional pricing tower and terminal settlement proved below.

Cohen–Reisinger–Wang [20] derive an HJM-type drift restriction and give hard constraints preserving a statically admissible factor domain. Their finite-factor implementation also constructs factors to reduce violations of the dynamic restriction, while Remark 2 explicitly distinguishes a rolling liquid lattice from fixed contracts. It would therefore be inaccurate to call their theory merely a penalty method. The comparison here is with a construction whose common conditional-expectation law makes the full stated family of prices martingales, rather than with a numerical objective that only reduces a drift residual.

Further foundations are Dupire's pricing identity [21], the DLV construction of Buehler–Ryskin [22], Andreasen–Huge's interpolation scheme [23], and martingale coupling in convex order [24, 25]. Deep hedging [2] motivates the joint simulation objective. Utility-based drift removal [27] provides a complementary diagnostic and measure-selection tool, but cannot create an equivalent martingale measure when the required span or support conditions fail.

1.3 Contributions, scope and organisation

The first result is the Wiese–Wissel drift equivalence in Section 2. Section 3 explains why a finite-grid drift identity leaves both the front boundary and a nonlinear interpolation open. Section 4 gives the smooth product theorem and an explicit all-maturity example. Section 5 extends the smoothing factor to time-varying and stochastic volatility, with a complete generator proof. Section 6 proves a local exact-fit result and specifies what common-node agreement means. Section 7 derives the continuous covariance drift and its relation to Carmona–Nadtochiy. Section 8 adds the Bates representation and nonrepresentation results, exact residual diagnostics, terminal rigidity theorems and direct-calibration alternatives. Sections 9–12 complete the front, regridding, physical-measure and recalibration programme. Section 13 states what numerical tests can and cannot certify.

The scope is deliberately precise. We prove absence of arbitrage for every finite portfolio of the constructed spot and option family under the usual admissibility restriction on trading strategies. We do not assert a large-market theorem for arbitrary infinite portfolios. We do not prove that every learned DLV volatility field has a global solution, that every historical surface can be represented with a fixed number of Black components, or that a finite numerical sweep establishes a universal existence claim. No new empirical performance results are asserted. Section 8 reports findings from the supplied Bates analyses with their provenance and unresolved audit limitations explicitly stated.

2 Reminder of the finite-grid Wiese–Wissel drift condition

2.1 Contracts, butterflies and the discrete Dupire identity

We work at zero interest and dividend rates on a finite horizon. Equivalently, an appropriate deterministic carry normalisation may first be applied. All martingale statements refer to a common measure $\mathbb{Q}$ and the market filtration. Fix cash strikes

$$0 < K_0 < K_1 < \cdots < K_{N+1}$$

and calendar expiries $T_1 < \cdots < T_M$. On a trading slab $[T_{r-1}, T_r)$ write $C_n^m(t) = C_t(T_m, K_n)$. A non-front maturity has $m > r$, so both T_{m-1} and T_m are live. Put $d_n = K_{n+1} - K_n$ and define interior butterflies

$$q_n^m = \frac{C_{n+1}^m - C_n^m}{d_n} - \frac{C_n^m - C_{n-1}^m}{d_{n-1}}, \qquad n = 1, \dots, N. \tag{2.1}$$

The q_n^m are dimensionless slope differences. If the conditional law is supported on the entire finite grid, augment them by

$$q_0^m = 1 + \frac{C_1^m - C_0^m}{d_0}, \qquad q_{N+1}^m = -\frac{C_{N+1}^m - C_N^m}{d_N},$$

with $C_0^m = S_t - K_0$, $C_{N+1}^m = 0$. Then

$$\sum_i q_i^m = 1, \quad \sum_i K_i q_i^m = S_t, \quad C_n^m = \sum_i q_i^m (K_i - K_n)^+. \tag{2.2}$$

When the true law has tails beyond the grid, (2.1) still defines valid interior butterflies, but the artificial boundary calls in (2.2) must not be substituted for the true tail prices.

For $q_n^m > 0$ define the backward discrete local variance

$$\ell_n^m = \frac{C_n^m - C_n^{m-1}}{\omega_n^m q_n^m}, \qquad \omega_n^m = \frac{K_n^2 (T_m - T_{m-1})}{K_{n+1} - K_{n-1}}. \tag{2.3}$$

This is an *exact coordinate identity* on a finite book. It is not an assertion that a finite difference equals a continuum derivative. Positivity of its numerator and denominator places the coordinates in the interior static-arbitrage domain. Degenerate zero-spread or zero-butterfly states require a boundary representation rather than division by zero.

2.2 The covariance drift and its converse

Let all live calls and spot be continuous local martingales, and write

$$dC_n^m = \Lambda_n^m \cdot dW^Q, \quad dq_n^m = \psi_n^m \cdot dW^Q, \quad d\ell_n^m = \alpha_n^m \, \mathrm{d}t + \beta_n^m \cdot dW^Q. \tag{2.4}$$

Theorem 2.1 (Wiese density-coordinate drift)**.** *For every positive interior non-front coordinate,*

$$\boxed{\alpha_n^m = -\frac{\beta_n^m \cdot \psi_n^m}{q_n^m}.} \tag{2.5}$$

The volatility identity is

$$\Lambda_n^m - \Lambda_n^{m-1} = \omega_n^m (q_n^m \beta_n^m + \ell_n^m \psi_n^m). \tag{2.6}$$

Conversely, suppose a positive DLV decoder satisfies (2.3), the previous-expiry calls and boundary calls are local martingales, and its coefficients satisfy (2.5). Then the decoded current-expiry calls are local martingales, up to exit from the admissible coordinate domain.

Proof. The product rule applied to $C_n^m - C_n^{m-1} = \omega_n^m \ell_n^m q_n^m$ gives drift $\omega_n^m(q_n^m \alpha_n^m + \beta_n^m \cdot \psi_n^m)$ and diffusion (2.6). This proves necessity.

For sufficiency let u_n be the drift of the decoded current call, and let B denote the slope-difference operator in (2.1). The butterfly drift is $(Bu)_n$. After inserting (2.5), the product drift identity becomes

$$u_n = \omega_n^m \ell_n^m (Bu)_n,$$

with zero boundary drift. The matrix $I - \mathrm{diag}(\omega^m \ell^m)B$ on the interior has positive diagonal, nonpositive off-diagonal entries and strict row diagonal dominance when augmented with identity boundary rows. It is invertible. Hence $u = 0$. Applying this argument successively gives the result for all back expiries. Local martingality becomes true martingality only with an additional integrability condition; Section 4 supplies one constructively. □

2.3 Exact equivalence with Wissel's non-front restriction

The precise predecessor is *Arbitrage-free market models for option prices*, Wissel's 2007 finite-option working paper [6]. The finite-grid reconciliation supplied in the Wiese research notes identifies its back-maturity equation (3.7). It is *not* equation (3.7) of the distinct Schweizer–Wissel 2008 multi-strike paper. To make the comparison independent of notation and equation numbering, we display the complete relation being reconciled.

Write v_n^m for Wissel's relative volatility of the local implied volatility x_n^m, and $\bar{u}_n^m$ for its drift under $\mathbb{Q}$:

$$\frac{dx_n^m}{x_n^m} = \bar{u}_n^m \, \mathrm{d}t + v_n^m \cdot dW^Q, \qquad \ell_n^m = (x_n^m)^2.$$

Define the relative calendar-spread volatility

$$R_n^m = \frac{\Lambda_n^m - \Lambda_n^{m-1}}{C_n^m - C_n^{m-1}}.$$

The Wissel back-maturity relation, expressed under the common martingale measure, is

$$\bar{u}_n^m = \left(-R_n^m + \frac{3}{2} v_n^m\right) \cdot v_n^m. \tag{2.7}$$

Theorem 2.2 (Exact non-front equivalence)**.** *On a common cash grid, for a common filtration and Brownian loading convention, with positive butterflies and calendar spreads,* (2.7) *and* (2.5) *are equivalent under the change of coordinates $\ell = x^2$. The equivalence concerns the same finite option prices and their drift restrictions, not a limiting approximation and not an identification of different front-expiry models.*

Proof. Itô's formula gives $\beta = 2\ell v$ and $\alpha/\ell = 2\bar{u} + \|v\|^2$. Equation (2.6) gives

$$R = \frac{\beta}{\ell} + \frac{\psi}{q} = 2v + \frac{\psi}{q}.$$

Consequently

$$2\bar{u} + \|v\|^2 = -2R \cdot v + 4\|v\|^2 = -2v \cdot \frac{\psi}{q} = -\frac{\beta}{\ell} \cdot \frac{\psi}{q},$$

which is (2.5). Every step is reversible for $\ell, q > 0$. □

The density formulation exposes the source of the singularity: the relative-density loading ψ/q. It does not remove that singularity. A strong-existence result can be transported through this coordinate map when the *same* front closure and coefficient hypotheses of the original existence theorem are imposed. There is no automatic extension to arbitrary fitted vol-of-DLV fields. The product constructions below provide their own existence mechanism instead.

2.4 Dual operators and the apparent equation count

For the compact-support case define the symmetric tridiagonal matrix B with interior rows (2.1), first row $(-1/d_0, 1/d_0, 0, \ldots)$ and last row $(\ldots, 1/d_N, -1/d_N)$. Then

$$\boldsymbol{q} = e_0 + BC, \quad D_m = \mathrm{diag}(0, \omega_1^m \ell_1^m, \ldots, \omega_N^m \ell_N^m, 0),$$

and the exact price and density recursions are

$$\begin{aligned} A_m^{\text{price}} C^m &= C^{m-1}, \qquad A_m^{\text{price}} = I - D_m B, \\ A_m^{\text{density}} \boldsymbol{q}^m &= \boldsymbol{q}^{m-1}, \quad A_m^{\text{density}} = I - B D_m = (A_m^{\text{price}})^\top. \end{aligned} \tag{2.8}$$

Indeed $B(I-DB) = (I-BD)B$ and $(I-BD)e_0 = e_0$. The affine term e_0 is essential; suppressing it obscures the boundary convention.

The N interior columns of B have zero mass and zero first moment. They are linearly independent: if their combination vanishes, extend the coefficients by zero at the two endpoints; the resulting vector lies in the kernel of B, which consists of constants, and is therefore zero. Their range is precisely the N-dimensional tangent space

$$\{z \in \mathbb{R}^{N+2} : \mathbf{1}^\top z = K^\top z = 0\}.$$

This explains why mass and mean remove two apparent density equations. It also separates the number of state coordinates from the number of Brownian factors.

3 What the finite-grid result leaves open

3.1 Linear interpolation preserves martingales, but not every payoff

For fixed cash strikes and a fixed expiry, deterministic linear interpolation of call-price martingales is again a martingale. This follows simply because a fixed linear combination of martingales is a martingale. The statement does not apply without extra terms to random interpolation weights, spot-relative regridding, implied-volatility interpolation or adaptive nonlinear fitting.

There is a separate settlement issue. At expiry the interpolated value is

$$(1-\theta)(S_T - K_i)^+ + \theta(S_T - K_{i+1})^+,$$

which need not equal $(S_T - K)^+$. For example, if $K_i = 80$, $K_{i+1} = 120$, $S_T = K = 100$, the chord settles at 10 and the actual call at 0. Linear interpolation therefore defines the price of a *portfolio of grid calls*. It defines the correct off-grid call only if the conditional terminal support has no mass in the relevant strike cell, as in a genuinely atomic terminal-support model. Martingale preservation and correct contract identification are distinct requirements.

3.2 Why a smooth decoder needs a drift restriction

If Z is a diffusion and $H(t, Z; T, K)$ a smooth decoder, then at fixed (T, K)

$$\mathrm{drift}_Q H = \partial_t H + DH\, b^Q + \tfrac{1}{2} D^2 H : (\Sigma\Sigma^\top). \tag{3.1}$$

An arbitrage-free cross-section does not make (3.1) vanish. The Hessian term is generally nonzero. A fit made independently after every simulated step may introduce further reset drift. Even if the decoder is exact at every protected node, its off-grid generator residual must be checked or eliminated structurally.

3.3 The front expiry is a terminal-boundary problem

For $m = r$ the predecessor in a front Dupire relation is the intrinsic payoff rather than another live option. For a continuous martingale spot, Tanaka's formula includes

$$d(S_t - K)^+ = \mathbf{1}_{\{S_t > K\}} dS_t + \tfrac{1}{2} dL_t^K(S).$$

The local-time term and the shrinking time interval prevent a mechanical reuse of the back-maturity derivation. In addition, each density must reach a terminal codebook compatible with the realised spot. A front marginal and a simulation rule for spot chosen independently need not have this compatibility. Section 9 constructs it explicitly.

4 A smooth SANOS product market

4.1 A coherent latent law and an independent Black martingale

Let X be a positive bounded càdlàg true martingale on $[0, T_*]$, adapted to $\mathcal{F}^X$. Let W^Y be independent of the entire X system under $\mathbb{Q}$. For deterministic σ_Y with $\int_0^{T_*} \sigma_Y^2(u)\,\mathrm{d}u < \infty$, set

$$Y_t = \exp\left(\int_0^t \sigma_Y(u)\,\mathrm{d}W_u^Y - \tfrac{1}{2}\int_0^t \sigma_Y^2(u)\,\mathrm{d}u\right), \quad S_t = X_t Y_t, \quad w(t,T) = \int_t^T \sigma_Y^2(u)\,\mathrm{d}u. \tag{4.1}$$

We use the joint filtration $\mathcal{F}_t = \mathcal{F}_t^X \vee \mathcal{F}_t^Y$. Define

$$\mathcal{B}(a, K, w) = \mathbb{E}[(a\exp(-w/2 + \sqrt{w}Z) - K)^+], \qquad Z \sim N(0,1), \tag{4.2}$$

with $\mathcal{B}(a, K, 0) = (a - K)^+$. For $w > 0$,

$$\mathcal{B} = a\Phi(d_1) - K\Phi(d_2), \quad d_1 = \frac{\log(a/K) + w/2}{\sqrt{w}}, \quad d_2 = d_1 - \sqrt{w}, \qquad \mathcal{B}_w = \tfrac{1}{2}a^2\mathcal{B}_{aa}.$$

If X_{T_m} takes values in a finite positive anchor set $\{x_i\}_{i=0}^L$, write

$$q_i^m(t) = \mathbb{Q}(X_{T_m} = x_i \mid \mathcal{F}_t^X), \qquad C_t(T_m, K) = \sum_{i=0}^{L} q_i^m(t)\mathcal{B}(x_i Y_t, K, w(t, T_m)). \tag{4.3}$$

These q^m are true *conditional atom probabilities*; they need not equal the butterflies of the *smoothed* call curve. The latent atom grid x and the protected observed strike grid K need not coincide.

Theorem 4.1 (Smooth product market and settlement)**.** *Under the preceding assumptions, S is a true $\mathbb{Q}$-martingale and*

$$C_t(T, K) = \mathbb{E}_Q[(S_T - K)^+ \mid \mathcal{F}_t] = \int \mathcal{B}(xY_t, K, w(t,T))\,\mathbb{Q}(X_T \in \mathrm{d}x \mid \mathcal{F}_t^X) \tag{4.4}$$

for every maturity $T \le T_*$. The finite formula (4.3) holds at every maturity with finite latent support. These prices:

- *(i) are true martingales in trading time for every fixed surviving contract;*
- *(ii) are decreasing and convex in K, satisfy the call bounds and put–call parity, and are increasing in maturity at fixed trading time;*
- *(iii) are C^∞ in $K > 0$ with strictly positive density when $w(t,T) > 0$;*
- *(iv) settle at $(S_T - K)^+$ for every strike, on and between any quote nodes.*

Consequently every finite submarket has a common martingale measure and is free of static and dynamic arbitrage under the standard admissibility conditions.

Proof. Independence and the martingale properties give $\mathbb{E}[X_T Y_T \mid \mathcal{F}_t] = X_t Y_t$. Conditioning first on X_T, the independent increment Y_T/Y_t is lognormal with mean one and total variance $w(t,T)$, proving (4.4). Calls are conditional expectations of integrable terminal payoffs bounded by S_T. The tower property yields true martingality. Conditional Jensen gives $C_t(T_2,K) \geq C_t(T_1,K)$ for $T_2 \geq T_1$. Strike monotonicity, convexity, bounds and parity follow directly from the payoff. The Black density is

$$\partial_{KK}\mathcal{B}(a,K,w) = \frac{1}{K\sqrt{2\pi w}} \exp\left[-\frac{(\log(K/a)+w/2)^2}{2w}\right] > 0. \tag{4.5}$$

On compact positive strike sets and away from $w=0$, bounded positive X allows differentiation under the conditional integral. This yields the smoothness assertion. At expiry the conditional expectation is the payoff itself. In the finite formula, $q_i^m(T_m) = \mathbf{1}_{\{X_{T_m}=x_i\}}$ and $w(T_m,T_m)=0$, so the same conclusion is immediate. Martingale prices rule out admissible arbitrage in any finite traded subfamily. □

4.2 The exact cancellation between nodes

Assume first that the conditional probabilities have continuous dynamics $dq_i^m = \psi_i^m \cdot dW^X$. Put $a_i = x_i Y_t$. At fixed T_m, K,

$$d\mathcal{B}(a_i,K,w) = a_i\sigma_Y \mathcal{B}_a\, dW^Y + \sigma_Y^2\left(-\mathcal{B}_w + \tfrac{1}{2}a_i^2\mathcal{B}_{aa}\right)dt.$$

The drift cancels by the Black heat equation. Independence makes the cross-variation with q_i^m zero. Hence

$$dC_t(T_m,K) = \left(\sum_i \mathcal{B}_i\psi_i^m\right)\cdot dW^X + \sigma_Y\left(\sum_i q_i^m a_i \mathcal{B}_{a,i}\right)dW^Y. \tag{4.6}$$

This is the required drift cancellation *at every strike*, rather than only at calibration nodes. If q has jumps, the same product rule applies with its compensated martingale part: the Black kernel is continuous and the price is linear in q.

If independence is removed, a continuous cross term appears:

$$\operatorname{drift}_Q C = \sum_i a_i \mathcal{B}_{a,i}\,\frac{d\langle q_i^m, \log Y\rangle_t}{dt}, \tag{4.7}$$

in addition to any kernel or probability drift. Thus independence is a substantive construction choice. Vanishing cross-variation can suffice for the local calculation, but by itself is not a proof of the conditional factorisation in (4.4).

The latent X is a martingale *process*, not a drift. Its induced non-front DLV coordinates obey (2.5). The Black factor supplies a second martingale and a smoothing kernel; the product rule links the two. Section 5 proves the corresponding extension beyond deterministic Black volatility. This distinction avoids interpreting a local-variance drift coefficient as an underlying factor.

4.3 An explicit finite-mixture construction for every maturity

To make existence and the all-maturity claim concrete, choose $0 < x_0 < \cdots < x_L$ and a finite-state Markov chain I_t with a bounded deterministic generator $A(t)$ satisfying

$$A_{ij}(t) \geq 0 \ (i \neq j), \qquad A(t)\mathbf{1} = 0, \qquad A(t)x = 0. \tag{4.8}$$

Then $X_t = x_{I_t}$ is a bounded positive martingale. One nontrivial choice has absorbing endpoints and, for $1 \leq i \leq L-1$,

$$A_{i,i-1} = \frac{\kappa_i(t)}{x_i - x_{i-1}}, \quad A_{i,i+1} = \frac{\kappa_i(t)}{x_{i+1}-x_i}, \quad A_{ii} = -A_{i,i-1} - A_{i,i+1}, \qquad \kappa_i \geq 0. \tag{4.9}$$

Let $P(t,T)$ be its transition matrix. For every $T > t$, the option price in state (i,y) is

$$H_i(t,y;T,K) = \sum_j P_{ij}(t,T)\mathcal{B}(x_j y, K, w(t,T)). \tag{4.10}$$

It satisfies the exact backward equation

$$\partial_t H_i + \tfrac{1}{2}\sigma_Y^2(t)y^2\partial_{yy}H_i + \sum_{j\neq i} A_{ij}(t)(H_j - H_i) = 0. \tag{4.11}$$

Indeed $\partial_t P = -A(t)P$ cancels the chain term, while the Black heat identity cancels the y and clock terms. Its terminal value is $(x_i y - K)^+$. Smooth coefficients give maturity regularity for $T > t$, and $w > 0$ gives strict strike smoothness. This is an explicit nontrivial, fixed finite-anchor SANOS-type surface with both no-arbitrage guarantees for *all* maturities.

For example, take $x = (80, 100, 120)$, $I_0 = 1$ and rates $A_{1,0} = A_{1,2} = r/2$, $A_{1,1} = -r$, with absorbing endpoints. Before the first jump,

$$H_1 = \tfrac{1-e^{-r(T-t)}}{2}\mathcal{B}(80y,K,w) + e^{-r(T-t)}\mathcal{B}(100y,K,w) + \tfrac{1-e^{-r(T-t)}}{2}\mathcal{B}(120y,K,w). \tag{4.12}$$

After a jump the price is the corresponding single Black call. Spot is not pinned to a fixed *cash* strike at expiry because Y_T remains random. This example has jumps in spot; a smooth option surface does not imply a continuous underlying. The diffusion-only Carmona–Nadtochiy boundary should not be applied to it without the jump extension. Also, an arbitrary calibrated martingale transition matrix need not be embeddable in such a chain: (4.8) is a distinct calibration restriction.

5 Time-varying and stochastic smoothing volatility

5.1 Deterministic time variation requires only the correct clock

Theorem 4.1 already permits any deterministic square-integrable $\sigma_Y(t)$, not merely a constant. The required update is

$$w(t,T) = \int_t^T \sigma_Y^2(u)\,\mathrm{d}u, \qquad \partial_t w(t,T) = -\sigma_Y^2(t).$$

This identity is precisely what cancels the Black gamma drift in (4.6). Replacing w by $\sigma_Y^2(t)(T-t)$ when σ_Y changes over time would usually be wrong: its additional time derivative has not been accounted for. Deterministic time variation of volatility is also different from *volatility of volatility*, which refers to a stochastic diffusion coefficient of the volatility state.

5.2 The general independent-kernel theorem

Let $\mathcal{G}$ be a filtration independent of the entire latent X system under Q, and let Y be a strictly positive continuous *true* $\mathcal{G}$-martingale, with $Y_0 = 1$. It may have stochastic volatility and additional internal state variables. For a fixed positive anchor x, define its conditional kernel

$$F_t^x(T,K) = \mathbb{E}_Q[(xY_T - K)^+ \mid \mathcal{G}_t]. \tag{5.1}$$

Theorem 5.1 (Product lift with stochastic smoothing volatility)**.** *For a bounded positive latent martingale X independent of the whole $\mathcal{G}$ system, $S = XY$ is a true martingale. At a model expiry with finite latent support,*

$$C_t(T_m,K) = \sum_i q_i^m(t)F_t^{x_i}(T_m,K) = \mathbb{E}_Q[(X_{T_m}Y_{T_m} - K)^+ \mid \mathcal{F}_t^X \vee \mathcal{G}_t]. \tag{5.2}$$

Every fixed-contract price is a true martingale with the correct payoff at maturity, and the whole family is statically arbitrage-free. If the conditional law of Y_T has a positive sufficiently differentiable density with locally dominated derivatives, the resulting call curve has the corresponding strict convexity and strike smoothness.

Proof. For $s \le t$, independence of the two filtered systems gives

$$\mathbb{E}[X_t Y_t \mid \mathcal{F}_s^X \vee \mathcal{G}_s] = \mathbb{E}[X_t \mid \mathcal{F}_s^X]\mathbb{E}[Y_t \mid \mathcal{G}_s] = X_s Y_s.$$

Integrability follows from bounded X and integrable Y. Conditioning the payoff on the finite-valued X_{T_m} and using the same independence gives (5.2). Its tower property, payoff limit, strike shape and calendar ordering follow exactly as in Theorem 4.1.

For the local product calculation in the Brownian smoothing models used below, each q_i^m is a martingale in the X system and each F^{x_i} is a continuous martingale in the independent smoothing system. Thus their continuous cross-variation vanishes, and there is no jump cross term. Consequently

$$d(q_i^m F^{x_i}) = F^{x_i}\, dq_i^m + q_{i,-}^m\, dF^{x_i}$$

has zero drift. The conditional-expectation representation upgrades this local calculation to true martingality. In a more general smoothing filtration, the kernels can themselves jump even if Y is continuous; retain the product jump terms, and use the conditional-expectation proof, which remains valid. If $f_t^Y(T, y)$ is the conditional density of Y_T, then

$$\partial_{KK} C_t(T_m, K) = \sum_i q_i^m(t) \frac{1}{x_i} f_t^Y(T_m, K/x_i). \tag{5.3}$$

Positivity and differentiation under this expression prove the smoothness assertion under the stated domination conditions. □

Independence is a sufficient structural assumption, not a necessary condition for every possible product martingale. In a continuous model,

$$d(XY) = Y\, dX + X\, dY + d\langle X, Y\rangle.$$

Zero covariation and suitable integrability can preserve a product martingale even without independence. They do not automatically imply the separated kernel formula (5.2). If smoothing volatility depends on the X state, the correct joint conditional pricing problem must be solved; the independent-kernel theorem cannot simply be reused.

The modification of the proof is therefore precise: the Black heat equation is replaced by the backward equation, or conditional-martingale property, of the *actual* stochastic-volatility kernel. The finite outer sum remains. Its individual terms are no longer generally single Black calls, so this is a generalised SANOS-type kernel family rather than the unchanged finite Black-mixture parametrisation.

5.3 A Markov stochastic-volatility kernel and its full Itô drift

Consider, within the independent smoothing system,

$$dY_t = Y_t \sigma(t, V_t) dB_t^1, \tag{5.4}$$

$$dV_t = b(t, V_t)dt + a(t, V_t)(\rho\, dB_t^1 + \sqrt{1-\rho^2}\, dB_t^2), \qquad |\rho| \le 1, \tag{5.5}$$

where B^1, B^2 are independent Brownian motions and the system has a nonexplosive solution. Independence from X allows leverage *inside* the smoothing system: ρ need not be zero. Assume Y is a true martingale. A useful sufficient condition on each finite horizon is

$$\mathbb{E}_Q \exp\left(\tfrac{1}{2}\int_0^{T_*} \sigma^2(u, V_u)\, \mathrm{d}u\right) < \infty; \tag{5.6}$$

in particular, bounded σ suffices. Square integrability of σ only pathwise defines a positive local martingale; without more it does not ensure a true martingale.

For a fixed x, T, K, put

$$f^x(t, y, v; T, K) = \mathbb{E}_Q[(xY_T - K)^+ \mid Y_t = y, V_t = v].$$

Where this function is classical, its backward equation is

$$\boxed{\partial_t f^x + \tfrac{1}{2}y^2\sigma^2 f^x_{yy} + b f^x_v + \tfrac{1}{2}a^2 f^x_{vv} + \rho y\sigma a f^x_{yv} = 0, \quad f^x(T, y, v; T, K) = (xy - K)^+.} \tag{5.7}$$

Itô's formula now gives the explicit zero-drift price dynamics

$$\begin{aligned} dC = &\left(\sum_i f^{x_i}\psi^m_i\right) \cdot dW^X \\ &+ \sum_i q^m_i (Y\sigma f^{x_i}_y + \rho a f^{x_i}_v) dB^1 + \sum_i q^m_i a\sqrt{1-\rho^2} f^{x_i}_v \, dB^2. \end{aligned} \tag{5.8}$$

The v-drift, volatility-of-volatility gamma and mixed derivative terms in (5.7) are the terms missing from a frozen-volatility Black overlay. Simply inserting the current V_t into $\mathcal{B}(xY_t, K, \sigma^2(t, V_t)(T-t))$ will not generally satisfy this equation.

If the smoothing state is multidimensional, replace the v terms by its full drift and covariance generator, including all cross derivatives with y. If volatility of volatility itself is stochastic, include its state in this generator. Theorem 5.1 is unchanged, provided the whole enlarged smoothing system stays independent of X and its Y factor remains a true martingale.

5.4 When the kernel is a mixture of Black kernels

In the uncorrelated autonomous-volatility case, the future volatility path is independent of the return Brownian motion. Set $A_{t,T} = \int_t^T \sigma^2(u, V_u)\,\mathrm{d}u$. Conditioning first on that path gives

$$f^x(t, y, v; T, K) = \mathbb{E}_Q[\mathcal{B}(xy, K, A_{t,T}) \mid V_t = v]. \tag{5.9}$$

This is the familiar stochastic-volatility mixing principle associated with Hull–White [26]. It is an expectation over the full future integrated variance, not the Black price at its mean: generally

$$\mathbb{E}[\mathcal{B}(xy, K, A_{t,T}) \mid \mathcal{G}_t] \neq \mathcal{B}(xy, K, \mathbb{E}[A_{t,T} \mid \mathcal{G}_t]).$$

If $A_{t,T} < \infty$ almost surely, independence from the return Brownian motion also proves $\mathbb{E}[Y_T/Y_t \mid \mathcal{G}_t] = 1$ directly by conditional Gaussian integration. This can establish true martingality even where a particular sufficient exponential-moment criterion is unavailable.

With leverage, write $dB^1 = \rho\, dB^V + \sqrt{1-\rho^2}\, dB^\perp$, with V autonomous and driven by B^V. Conditional on its driving path, define

$$M_{t,T} = \exp\left(\rho \int_t^T \sigma(u, V_u)\,\mathrm{d}B^V_u - \tfrac{1}{2}\rho^2 A_{t,T}\right).$$

The corresponding representation is

$$f^x(t, y, v; T, K) = \mathbb{E}_Q[\mathcal{B}(xyM_{t,T}, K, (1-\rho^2)A_{t,T}) \mid V_t = v]. \tag{5.10}$$

The random forward multiplier is essential. A variance-only mixture such as (5.9) is generally incorrect when $\rho \neq 0$. The representation assumes that the remaining Brownian motion $B^\perp$ is independent of the autonomous volatility subsystem; the general backward equation covers cases without this convenient decomposition.

5.5 A fully specified smooth example with stochastic volatility of volatility

Choose $0 < \underline{\sigma} < \overline{\sigma} < \infty$, $|\rho| < 1$, and independent Brownian motions $B^V, B^\perp, B^U$, all independent of X. Set

$$\begin{aligned} dU_t &= -\kappa_U U_t dt + \zeta\, dB_t^U, \\ dV_t &= \kappa_V(\theta - V_t)dt + e^{U_t}\, dB_t^V, \\ \sigma(v) &= \underline{\sigma} + \frac{\overline{\sigma} - \underline{\sigma}}{1 + e^{-v}}, \\ \frac{dY_t}{Y_t} &= \sigma(V_t)(\rho\, dB_t^V + \sqrt{1-\rho^2}\, dB_t^\perp). \end{aligned} \tag{5.11}$$

Take $\kappa_U, \kappa_V > 0$, $\zeta \geq 0$ and finite initial states. The OU process U has continuous finite paths on each compact horizon; therefore e^U is locally square integrable and the linear V equation has a nonexplosive solution. Bounded σ makes the stochastic exponential Y a true martingale. The instantaneous volatility process $\sigma(V_t)$ has diffusion coefficient $\sigma'(V_t)e^{U_t}$, so its volatility of volatility is genuinely stochastic.

Conditional on (B^V, B^U) after time t, the residual Black variance satisfies

$$0 < (1-\rho^2)\underline{\sigma}^2(T-t) \leq (1-\rho^2)A_{t,T} \leq (1-\rho^2)\overline{\sigma}^2(T-t).$$

Formula (5.10), now conditional on (V_t, U_t), therefore has a strictly positive C^∞ strike density for every $T > t$. On compact positive strike sets, the kernel-density derivatives are uniformly bounded over the random forward multiplier and this bounded variance interval, justifying differentiation. Theorem 5.1 consequently provides a smooth, statically and dynamically arbitrage-free product market with stochastic smoothing volatility *and* stochastic volatility of volatility.

The generator used to price its kernel adds $-\kappa_U u\,\partial_u + \frac{1}{2}\zeta^2\partial_{uu}$ to (5.7), sets $a = e^u$, and retains the yv cross term $\rho y\sigma(v)e^u\partial_{yv}$. There are no yu or vu cross terms because B^U is independent. Every term is thus specified. No universal upper bound on e^{U_t} is needed for the true-martingale argument in this example; bounded *return* volatility is what supplies it.

5.6 Consequences for the DLV drift, node fitting and recalibration

The X-only Wiese–Wissel restriction is unchanged. The observed smoothed calls use the enlarged volatility vector (5.8); their induced finite and continuous DLV coordinates therefore satisfy the same covariance identities with that full vector. If the enlarged system has jumps, its predictable jump correction must also be included. There is no extra free off-grid drift to be chosen independently of these kernel dynamics.

Exact node calibration remains linear in the outer weights for a *fixed conditional kernel*, although evaluating its columns can be more expensive than evaluating Black. The local-fit argument of Theorem 6.2 extends to a scaled family $dY^\eta/Y^\eta = \eta\sigma(V)dB^1$ with bounded σ: $Y_T^\eta \to 1$ in L^1 as $\eta \to 0$, and the Lipschitz call payoff makes the kernel matrix converge to its intrinsic-value matrix. The same invertibility, positivity and finite strict-calendar argument applies. An exact expectation kernel gives an exact theorem; quadrature or a neural kernel approximation introduces a numerical residual that must be audited.

Recalibration may update smoothing-state parameters only through an admissible state evolution or a price-preserving reset. An independently fitted stochastic-volatility kernel at each date is no more dynamically guaranteed than an independently fitted Black variance at each date. The CRC programme in Section 12 therefore carries over with the kernel state replacing the single deterministic variance clock.

6 Exact common-node agreement and calibration freedom

6.1 Why a frozen DLV distribution cannot simply be smoothed

Proposition 6.1 (Positive Black smoothing changes a frozen atomic book)**.** *For $a, K, w > 0$, $\mathcal{B}(a, K, w) > (a - K)^+$. Consequently a positive mixture with fixed weights and anchors cannot preserve its unsmoothed call prices at any positive strike when $w > 0$.*

Proof. The nondegenerate lognormal variable in (4.2) has mean a and positive probability on either side of every $K > 0$. The call payoff is convex and its kink is crossed with positive probability, so conditional Jensen is strict. Summing the strict inequalities with nonnegative weights of total mass one proves the claim. □

This observation is essential for the meaning of "agrees exactly with DLV on common nodes". The construction does not preserve a pre-existing atomic grid by smoothing its unchanged probabilities. There are instead two valid statements:

1. *Jointly calibrated common nodes.* Choose the smoothing clock and latent probabilities together to reproduce protected quote prices. Evolve a coherent conditional law from that initial state. Its subsequent grid and off-grid prices are parts of the same model.
2. *Induced DLV coordinates.* Restrict the constructed smooth call field to a cash grid and define its DLV coordinates through (2.1)–(2.3). These coordinates reconstruct the same protected calls and satisfy the Wiese–Wissel condition. They are generally different from the latent unsmoothed X-DLVs.

Neither statement says that an arbitrary independently prescribed future DLV path has a finite Black-mixture extension with unchanged dynamics.

6.2 A local exact-fit theorem with positive smoothing

At a calibration date, normalise $Y_0 = 1$ and let the latent anchors also define the protected *interior* strikes: $x_0 < K_1 = x_1 < \cdots < K_{L-1} = x_{L-1} < x_L$. There are $L + 1$ unknown probabilities per expiry and $L - 1$ protected calls, plus mass and mean. For fixed w_m the calibration equations are linear:

$$\sum_i q_i^m = 1, \quad \sum_i x_i q_i^m = S_0, \quad \sum_i q_i^m \mathcal{B}(x_i, K_n, w_m) = c_n^m, \qquad n = 1, \dots, L-1. \tag{6.1}$$

Require $q^m \geq 0$ and latent convex order $q^{m-1} \preceq_{\mathrm{cx}} q^m$. Convex order on a common finite support is equivalent to equality of means and the call inequalities at the interior anchors, so these additional constraints are linear too.

Theorem 6.2 (A nonzero smoothing interval with exact protected quotes)**.** *Suppose the target book in (6.1) at $w_m = 0$ has strictly positive probability vectors $q^{m,0}$ on the chosen support, a common mean S_0, and strictly positive latent calendar spreads at every interior anchor between successive expiries. Choose a deterministic clock V_0 strictly increasing at the model expiries and set $w_m = \eta^2 V_0(T_m)$. There is $\eta_* > 0$ such that for every $0 < \eta < \eta_*$:*

(i) (6.1) has a unique solution on this fixed anchor set;
(ii) all its weights remain strictly positive and its latent marginals remain in convex order;
(iii) these marginals admit a bounded martingale coupling, and hence a product model with smooth all-strike prices that matches the protected initial quotes exactly and settles correctly at the model expiries.

The theorem provides a family of exact finite-quote fits; it does not assert a unique smoothing parameter.

Proof. Let $M(w)$ be the square matrix whose rows are mass, first moment and the $L-1$ Black calls in (6.1). At $w=0$, mass, mean and all interior call values determine the finite law uniquely by successive slope differences and the two boundary masses. Thus $M(0)$ is invertible. Its entries are continuous in w, including at zero, so $M(w_m)^{-1}$ exists for sufficiently small η. The solution $q^m(\eta) = M(w_m)^{-1}(1, S_0, c^m)^\top$ is continuous at $q^{m,0}$. Since only finitely many positivity and strict interior calendar inequalities are imposed, they remain valid on a common interval $(0, \eta_*)$.

The finite convex-order coupling theorem [24] supplies transition matrices P_m with $P_m\mathbf{1} = \mathbf{1}$, $P_m x = x$, and $q^{m-1}P_m = q^m$; the first marginal is coupled to the deterministic value S_0. Section 9.2 constructs a continuous bounded martingale with these date marginals. Apply Theorem 4.1 with the independent clock $\eta^2 V_0$. Equation (6.1) gives the exact initial match. □

The strict hypotheses matter. For a boundary book, such as a zero-price upper-tail call, positive Black smoothing can make exact fit impossible on that prescribed support. In particular, compact-support boundary identities $C(x_L) = 0$ and $C(x_0) = S_0 - x_0$ are not additional protected quotes in Theorem 6.2. The smoothed law has tails beyond the latent anchors. A realistic calibration uses interior quotes with explicit tail modelling, or works within bid–ask bounds.

For a larger or irregular quote set, exact feasibility is the convex-hull condition that the target vector lies in the convex hull of the kernel columns, with the mean and calendar restrictions included. A least-squares residual, a positivity failure, poor conditioning and a calendar inconsistency are different outcomes and must be reported separately. Increasing the number of anchors can improve feasibility but does not prove it for every surface.

6.3 What is inherited on the common nodes

Corollary 6.3 (Exact price agreement and induced DLV drift)**.** *Let C be the product surface and choose any fixed finite quote grid. On every live slab where the interior butterflies and calendar spreads are positive, define q^{obs} and ℓ^{obs} by (2.1)–(2.3), using the actual product prices at the neighbouring and boundary strikes. Then the observed DLV identity holds exactly on that grid. For continuous dynamics its coefficients obey (2.5); for jumps they obey (7.8) below. At each jointly calibrated reset the protected quotes are exactly preserved whenever the inverse problem is feasible and solved exactly.*

Proof. Theorem 4.1 makes every call a true martingale. The butterfly operator is fixed and linear, so the interior observed butterflies are martingales. The proof of Theorem 2.1 then applies to their defining ratio. The last statement is the calibration equality itself. □

The latent atom weights q^X and the observed butterflies q^{obs} are different codebooks for different roles. Treating them as the same vector would incorrectly reintroduce Proposition 6.1. Likewise, a finite butterfly is not the pointwise continuous density. "Consistent drift restrictions" means the same price-martingale identity expressed through the appropriate operator, not numerical equality of all discrete and continuous local variances.

6.4 How large, how variable, and how identifiable is smoothing?

For the deterministic Black clock there is no universal finite upper bound on σ_Y: square integrability on each finite horizon suffices for the true exponential martingale. Smoothness for each live maturity requires $w(t,T) > 0$. Calibration imposes further bounds. Convexity in the first Black argument gives

$$C_t(T,K) \geq \mathcal{B}(S_t, K, w(t,T)). \tag{6.2}$$

Consequently $w(t,T)$ cannot exceed the total Black implied variance of any exactly fitted call at that maturity. This necessary condition is not sufficient for simultaneous mixture feasibility.

Finite quotes generally do not identify η uniquely; Theorem 6.2 demonstrates an interval explicitly. A choice can be made by out-of-sample interpolation error, density regularity, numerical

conditioning and historical transition fit, subject to exact no-arbitrage constraints. All positive values in the theorem already give C^∞ strike smoothness, so requiring merely C^2 does not select one value.

A deterministic time-dependent $\eta(t)$ is harmless when interpreted through one additive clock, for example $w(t,T) = \int_t^T \eta^2(u)\sigma_0^2(u)\,\mathrm{d}u$. An independently recalibrated random scalar η_t inserted into $\eta_t^2(T-t)$ is a different model: its drift, variance and cross-variations enter (3.1). It is not covered by the Black cancellation. One can instead model an independent stochastic-volatility factor Y and use its *actual conditional pricing kernel*; this retains the conditional-expectation proof but generally leaves the single-Black-kernel family. At discrete parameter resets a new deterministic future clock is also possible if the price-preserving stitching conditions in Section 12 hold.

7 The drift condition between nodes

7.1 A common identity for finite and continuous codebooks

Let $C_t(T,K)$ be a sufficiently regular smooth surface. For fixed calendar maturity and cash strike define

$$p_t(T,K) = \partial_{KK} C_t(T,K) > 0, \qquad \ell_t(T,K) = \frac{2\partial_T C_t(T,K)}{K^2 p_t(T,K)}. \tag{7.1}$$

Here p is a density, whereas the q_n in (2.1) are slope differences. Assume derivatives can be interchanged with conditional expectation and the semimartingale decomposition, locally away from expiry and any parameter singularity. Sufficient local domination and stochastic-Fubini conditions are part of this assumption; pointwise smoothness alone is not enough.

Theorem 7.1 (Continuous density-coordinate restriction)**.** *Suppose $C_t(T,K)$ is a continuous local-martingale field and the preceding regularity holds. If*

$$dp = \psi \cdot dW^Q, \qquad d\ell = \alpha\,\mathrm{d}t + \beta \cdot dW^Q,$$

then

$$\boxed{\alpha_t(T,K) = -\frac{\beta_t(T,K)\cdot\psi_t(T,K)}{p_t(T,K)}.} \tag{7.2}$$

The same identity holds with the maturity derivative replaced by a fixed difference $[C(T_m,K) - C(T_{m-1},K)]/\Delta T_m$ and with

$$\ell^{m,\mathrm{sec}}(K) = \frac{2[C(T_m,K) - C(T_{m-1},K)]}{K^2 \Delta T_m\, p(T_m,K)}. \tag{7.3}$$

Thus an exact between-strike drift relation is available even if only model expiries are represented by finite mixtures.

Proof. The left side of $\partial_T C = \frac{1}{2}K^2\ell p$ is a local martingale under the differentiation assumptions. The product on the right has drift $\frac{1}{2}K^2(p\alpha + \beta\cdot\psi)$, proving (7.2). A fixed difference of live call martingales is a martingale, so the identical argument proves the secant version. □

This proof also clarifies the exact finite-grid comparison. Replacing the linear curvature operator ∂_{KK} by the butterfly operator and the maturity derivative by the discrete calendar operator yields Theorem 2.1. Sampling a smooth oracle is enough to define the *price* grid, but its DLV coordinates must then be *recomputed from the finite ratios*; they are not obtained by sampling the oracle's continuous local variance.

More precisely, for the triangular hat function h_n supported on $[K_{n-1}, K_{n+1}]$, with $h_n(K_n) = 1$, integration by parts gives

$$q_n^m = \int_{K_{n-1}}^{K_{n+1}} h_n(K) p(T_m,K)\,\mathrm{d}K. \tag{7.4}$$

This is an exact weighted density integral. It explains both the relationship and the nonidentity between the two codebooks. No mesh-convergence theorem is needed for the finite drift identity, and coarse strike spacing cannot excuse failure of that identity when the same price map and generator are used consistently.

7.2 Explicit smooth drift coefficients for the product

In the continuous X version of (4.3), let $\nu(T,K)$ denote the entire call-volatility vector in (4.6). Wherever maturity differentiation is valid, set

$$\psi = \partial_{KK}\nu, \qquad \xi = \frac{2}{K^2}\partial_T\nu.$$

The martingale numerator $N = 2\partial_T C/K^2$ satisfies $dN = \xi \cdot dW$, and $\ell = N/p$. Therefore the exact coefficients are

$$\boxed{\beta = \frac{\xi - \ell\psi}{p}, \qquad \alpha = \frac{\ell\|\psi\|^2 - \xi\cdot\psi}{p^2} = -\frac{\beta\cdot\psi}{p}.} \tag{7.5}$$

For the secant codebook replace ξ by $2[\nu(T_m,K) - \nu(T_{m-1},K)]/(K^2\Delta T_m)$. These formulas require no Bates oracle. They follow from the actual SANOS-type decoder and its coherent dynamics.

For example, write $g_i(K) = \partial_{KK}\mathcal{B}(x_iY_t, K, w)$ as in (4.5). Then

$$p = \sum_i q_i^m g_i, \qquad \psi^X = \sum_i g_i\psi_i^m, \qquad \psi^Y = \sigma_Y \sum_i q_i^m a_i \partial_a g_i.$$

Together with (4.6), this specifies all terms of (7.5). The absence of call drift is exact; the local variance itself generally has the nonzero covariance drift displayed above. The latent DLV drift is therefore not simply copied unchanged into the gaps.

7.3 Relation to Carmona–Nadtochiy's clock and closure

Introduce time to maturity $\tau = T - t$, log strike $z = \log K$, and

$$\widehat{C}_t(\tau,z) = C_t(t+\tau, e^z), \qquad D_z = \tfrac{1}{2}(\partial_{zz} - \partial_z), \qquad \widehat{a}_t^2(\tau,z) = \ell_t(t+\tau,e^z).$$

Then $D_z\widehat{C} = \frac{1}{2}K^2p$. If $d\widehat{a}^2 = \widehat{a}^2(\widehat{\alpha}\,\mathrm{d}t + \widehat{\beta}\cdot dW)$ at fixed (τ,z), Itô–Wentzell for the deterministic shift gives

$$\alpha = \ell(\widehat{\alpha} - \partial_\tau \log\widehat{a}^2), \qquad \beta = \ell\widehat{\beta}.$$

Hence (7.2) becomes

$$\widehat{\alpha} + \widehat{\beta}\cdot\frac{D_z\widehat{\nu}}{D_z\widehat{C}} = \partial_\tau \log\widehat{a}^2. \tag{7.6}$$

This is the Carmona–Nadtochiy drift relation [11]: Theorem 1, equation (8) and its first remark in their author-hosted May 2007 preprint; the published version uses different numbering. The transport term is essential when comparing a rolling surface with fixed contracts.

For a continuous underlying the accompanying spot-consistency condition is

$$\widehat{a}_t^2(0,\log S_t) = \frac{d\langle S\rangle_t}{S_t^2\,dt},$$

when the short-end limit exists in the required sense. There is also a volatility closure, obtained by taking the martingale coefficient in the Dupire identity:

$$\partial_T\nu = \tfrac{1}{2}K^2(\ell\,\partial_{KK}\nu + p\beta). \tag{7.7}$$

Thus the covariance restriction alone, imposed on unrelated fields (ℓ,p), is not a complete existence theorem. The product construction first supplies consistent calls; their induced fields satisfy the restrictions. Conversely, a proposed free field must also satisfy closure, settlement, regularity and integrability.

7.4 Jumps and non-diffusive latent models

Write the continuous martingale loadings of (ℓ, p) as (β, ψ) and their jump sizes for mark e as $(\delta\ell(e), \delta p(e))$, with compensator $\nu_t(de)dt$. If p and ℓp are local martingales and $p_- > 0$, the predictable product drift gives

$$\boxed{\alpha = -\frac{\beta \cdot \psi + \int \delta\ell(e)\delta p(e)\nu_t(de)}{p_-}.} \tag{7.8}$$

The same formula holds with a finite butterfly q in place of p. It applies in particular to the chain model of Section 4.3. The ratio $2C_T/(K^2 p)$ remains a valid positive surface coordinate in that model, but is not generally the instantaneous diffusion variance of its jumping underlying. Claiming the diffusion-only short-end condition for it would conflate two different statements.

8 Generalised Bates as an oracle: representation, rigidity and constructive alternatives

The two supplied analyses [28, 29] add several results to the product and drift theory above. Their central distinction is between reproducing a *specified* Bates market and constructing a coherent SANOS market of its own. This section incorporates the additional representation theorems, exact diagnostic identities, terminal obstructions and constructive calibration alternatives. Results already proved in Sections 4–7 are used rather than restated as new discoveries. The numerical findings below are attributed to the supplied analyses; no new simulation or code audit is claimed.

8.1 The reference model and the status of the numerical evidence

Here "Generalised Bates" denotes the fixed-characteristics reference used in those analyses: Heston variance with an independent finite-activity Gaussian log-jump extension, whose intensity and mark parameters may have a deterministic calendar schedule. On a live interval write

$$\begin{aligned} dV_u &= \kappa(\theta - V_u)\,\mathrm{d}u + \xi\sqrt{V_u}\,\mathrm{d}B_u, \\ d\log S_u &= -\tfrac{1}{2}V_u\,\mathrm{d}u + \sqrt{V_u}(\rho\,\mathrm{d}B_u + \sqrt{1-\rho^2}\,\mathrm{d}W_u) + \mathrm{d}J_u - k_J(u)\,\mathrm{d}u, \\ k_J(u) &= \int (e^z - 1)\nu_u(\mathrm{d}z). \end{aligned} \tag{8.1}$$

The Brownian motions B, W and the independent-increment jump process J are mutually independent. Assume the usual nonnegative, nonexplosive variance solution, finite positive-order jump moments on the finite horizon, and a true-martingale stock. The nonrepresentation results below use $V_t > 0$, $\xi > 0$, $|\rho| < 1$ and $T - t > 0$. Define

$$C_t^{\mathrm{B}}(T, K) = \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ \mid \mathcal{F}_t].$$

This reference is not the same object as a fully implemented, state-dependent Gambara–Teichmann CRC model. The supplied analyses explicitly describe the available reset inverse as a *restricted CRC approximation*. Exact claims about (8.1) cannot certify the approximate reset sequence. A random change of structural parameters or a jump-extension correction must be handled by the actual conditional law and the reset conditions of Section 12.

Table 1: Earlier numerical findings as reported in [28], and their evidential limits. These are not independently reproduced results of the present paper.

Reported diagnostic	Reported value or procedure	What it establishes, or leaves open
Finite reconstruction at zero smoothing	Protected-price error 6.9×10^{-18}.	Excellent finite-grid reconstruction; no conclusion about unobserved strikes or dynamic closure.
Selected smooth fit	At the report's $w = \eta^2$, $\eta = 0.05$: protected-price error 4.3×10^{-5}; nonzero off-grid errors.	An accurate approximation, not an exact full-surface Bates representation or a certified equality-constrained fit.
Finite DLV drift check	Jump-corrected residual 0.0261, attributed there to $O(\Delta K^2, \Delta T)$ truncation.	That explanation cannot apply to the exact coordinate identity (8.3). State differentiation, quadrature and inconsistent definitions must be investigated separately.
Selected-fit dynamic check	Some residuals were reused from an older moving-query reconstruction.	Prices, derivatives and residuals of different decoders do not establish the generator residual of the selected fit.
Oracle correction	$G^\star = C^{\mathrm{B}} - H^0$; the older H^0 reportedly missed protected quotes by up to 0.946.	$H^0 + G^\star = C^{\mathrm{B}}$ is an identity, not a finite SANOS construction. The correction is not grid-vanishing when H^0 misses the grid.
CRC reset	Restricted inverse approximation.	Reset errors require their own audit; the exact fixed-reference null and approximate CRC evolution must remain distinct.

Only the two supplied analyses, not the underlying Task 7 implementation or its machine-readable run outputs, support this table. The discrepancies identify reproducible questions, not a diagnosed faulty line of code. The value 0.0261 is quoted without imposing a new normalisation not present in the supplied analysis.

8.2 An exact diagnostic for projection and reconstruction

Let Z be the declared oracle or reconstruction state and let $\mathcal{A} = \partial_t + \mathcal{L}$ be its full fixed-contract generator. With the compensated-jump convention

$$dZ = b\,\mathrm{d}t + \Sigma\,\mathrm{d}W + \int j(e)\,\widetilde{N}(\mathrm{d}t, \mathrm{d}e),$$

the generator of a sufficiently regular price map is

$$\begin{aligned}\mathcal{A}H = \partial_t H + b \cdot \nabla H + \tfrac{1}{2}\operatorname{tr}(\Sigma\Sigma^\top D^2 H) \\ + \int \big[H(t, Z + j(e)) - H(t, Z) - \nabla H \cdot j(e)\big]\nu_t(\mathrm{d}e).\end{aligned} \tag{8.2}$$

The state drift and jump truncation must use this same convention. Define

$$\Gamma^c(f, g) = (\Sigma^\top \nabla f) \cdot (\Sigma^\top \nabla g), \qquad \mathcal{J}(f, g) = \int \delta f(e)\delta g(e)\nu_t(\mathrm{d}e).$$

Take *any* candidate finite call book, form its butterflies q and ratio ℓ exactly as in (2.1)–(2.3), and suppress the node indices. On a non-front live slab,

$$\boxed{q\mathcal{A}\ell + \Gamma^{c}(\ell, q) + \mathcal{J}(\ell, q) = \frac{\mathcal{A}(C^{m} - C^{m-1})}{\omega} - \ell\mathcal{A}q.} \tag{8.3}$$

Indeed, apply the generator product rule to the defining identity $\omega\ell q = C^{m} - C^{m-1}$. This proves (8.3) even when the candidate prices are not martingales. For exact oracle prices, both generators on the right vanish and the jump-extended Wiese–Wissel restriction follows. No limit in strike spacing or maturity spacing is taken.

Thus discretisation of the oracle means sampling *call prices*, then computing finite slope differences and ratios; it does not mean sampling Heston variance or a continuous Dupire coefficient. The reverse finite-decoder implication still needs its front and boundary assumptions. Furthermore, a projection of the full Bates state onto finitely many quotes need not be a closed Markov state: the identity alone supplies neither sufficient statistics nor an autonomous DLV transition law.

There is an equally useful continuous diagnostic. Put $R_C = \mathcal{A}H$, $p = H_{KK} > 0$ and $\ell^{c} = 2H_T/(K^2 p)$. If contract differentiation commutes with $\mathcal{A}$, then

$$\begin{aligned} &\mathcal{A}p = \partial_{KK} R_C, \\ &pR_\ell = \frac{2}{K^2}\partial_T R_C - \ell^{c}\partial_{KK} R_C, \qquad R_\ell = \mathcal{A}\ell^{c} + \frac{\Gamma^{c}(\ell^{c}, p) + \mathcal{J}(\ell^{c}, p)}{p}. \end{aligned} \tag{8.4}$$

This follows by applying the same product rule to $\ell^{c} p = 2H_T/K^2$. It connects call-space and density-coordinate residuals without equating them numerically. Density evolution, boundary information and settlement remain necessary; $R_\ell = 0$ alone is not a complete market theorem. With jumps, ℓ^{c} is a surface ratio, not the instantaneous diffusion variance of the Bates stock. The Carmona–Nadtochiy connection in Section 7 must retain these jump and boundary qualifications.

8.3 The finite-family obstruction and identification of a common width

At one state and expiry, a positive finite Black mixture has the form

$$H(K) = \sum_{j=1}^{N} \pi_j \mathcal{B}(a_j, K, w_j), \qquad \pi_j \geq 0, \quad \sum_j \pi_j = 1, \quad \sum_j \pi_j a_j = S_t, \tag{8.5}$$

with finite anchors $a_j \geq 0$ and finite variances $w_j \geq 0$. The common-width production subfamily imposes $w_j = w$; permitting different finite w_j does not remove the following obstruction.

Theorem 8.1 (Nondegenerate Bates has no exact finite Black-mixture representation)**.** *Under the nondegeneracy assumptions of Section 8.1, no mixture* (8.5) *equals $C_t^{\mathrm{B}}(T, K)$ for every $K > 0$. The anchors, weights and variances may depend on the current state, trading time and expiry; the assertion holds pointwise at every positive remaining maturity.*

Proof. The random variable represented by (8.5) has every positive-order moment finite:

$$M(p) = \sum_j \pi_j a_j^{p} \exp\{\tfrac{1}{2} w_j p(p-1)\} < \infty, \qquad p > 0.$$

For the Heston part, the affine moment coefficient obeys

$$D_p' = aD_p^2 + b_p D_p + c_p, \qquad D_p(0) = 0, \quad a = \tfrac{1}{2}\xi^2, \quad b_p = \rho\xi p - \kappa, \quad c_p = \tfrac{1}{2}p(p-1). \tag{8.6}$$

For sufficiently large p, $b_p^2 - 4ac_p = \xi^2(\rho^2 - 1)p^2 + O(p) < 0$. Its explosion time is

$$T^{*}(p) = \int_0^{\infty} \frac{\mathrm{d}u}{au^2 + b_p u + c_p} = \frac{2}{\sqrt{4ac_p - b_p^2}} \left[\frac{\pi}{2} - \arctan\frac{b_p}{\sqrt{4ac_p - b_p^2}}\right] = O(p^{-1}). \tag{8.7}$$

The affine moment criterion [30] therefore gives, at every fixed positive remaining maturity, an infinite moment of some sufficiently large finite order. The independent Gaussian compound-Poisson factor has a finite strictly positive moment at that order and cannot remove the explosion. Equality of all call prices determines equality of terminal probability laws, whereas these laws have incompatible moments. □

For the numerical parameters quoted in [28], $\kappa = 2$, $\xi = 0.35$ and $\rho = -0.7$, (8.7) gives $T^*(100) = 0.21559324\ldots$ years. Hence at a half-year horizon with positive current variance the reference has an infinite 100th moment. Every finite Black mixture has a finite one. This is a separating tail property, not a prediction of large errors at liquid strikes. Degenerate targets, such as deterministic variance with no jumps, fall outside the theorem and may be exactly Black.

In contrast, finitely many Bates quotes can satisfy Theorem 6.2. At a single expiry the explicit weights are $\pi^\eta = M(\eta^2)^{-1}(1, S_t, c_1, \ldots, c_n)^\top$ whenever the square kernel matrix is invertible and the weights are positive. With more columns, its null space supplies additional fitting freedom, subject to positivity and calendar constraints. The existence of a small positive interval does not certify a particular reported value such as $\eta = 0.05$.

Proposition 8.2 (A complete finite common-width curve identifies its width)**.** *If a positive-mean law is exactly a finite common-width Black mixture, its common total variance is determined by its full call curve. For any fixed reference scale $s_* > 0$,*

$$w = \lim_{p\to\infty} \frac{2\log \mathbb{E}[(S/s_*)^p]}{p^2}. \tag{8.8}$$

Proof. The logarithm of the moment equals $\frac{1}{2}wp(p-1) + \log\sum_j \pi_j(a_j/s_*)^p$. The second term is $O(p)$ for a finite set with at least one active positive anchor. Divide by $p^2/2$. The full call curve determines the law and hence its moments. □

Redundant components do not create a second common width. Thus finite-quote non-identification of η and full-curve identification within this subfamily are compatible. For the nondegenerate Bates target, there is no exact finite-mixture width to identify.

8.4 Why a common positive Gaussian floor cannot reproduce this oracle

The obstruction can be strengthened in a different direction. Even an infinite positive mixture with one common $w > 0$ would factor the log-price characteristic function as

$$\phi(u) = e^{-wu^2/2 - iwu/2}\psi(u), \qquad |\psi(u)| \le 1, \quad \text{so } |\phi(u)| \le e^{-wu^2/2}.$$

The reference (8.1), without an additional independent Brownian variance floor, does not satisfy that bound for any $w > 0$. To see this directly, set $\tau = T - t$, $b(u) = \kappa - i\rho\xi u$ and

$$d(u) = \sqrt{b(u)^2 + \xi^2(u^2 + iu)}, \qquad \Re d(u) > 0, \qquad g(u) = \frac{b(u) - d(u)}{b(u) + d(u)}.$$

The affine characteristic formula has coefficients

$$D(u,\tau) = \frac{b-d}{\xi^2}\frac{1 - e^{-d\tau}}{1 - ge^{-d\tau}},$$
$$F(u,\tau) = \frac{\kappa\theta}{\xi^2}\left[(b-d)\tau - 2\log\frac{1 - ge^{-d\tau}}{1-g}\right].$$

They solve the affine backward equation with zero terminal coefficients. Since $\Re d(u) = \xi\sqrt{1-\rho^2}|u| + O(1)$, the real part of $F + V_t D$ gives

$$\log|\phi(u)| = -c_\tau|u| + O(\log(2 + |u|)), \qquad c_\tau = \frac{\sqrt{1-\rho^2}}{\xi}(V_t + \kappa\theta\tau) > 0. \tag{8.9}$$

The finite-activity jump exponent has bounded real part and leaves this leading order unchanged. Thus $-2\log|\phi(u)|/u^2 \to 0$, contradicting a common positive Gaussian floor. The same argument excludes component variances all bounded below by one fixed $\varepsilon > 0$, since that common floor can be split off from every component. Adding an independent Black factor to the *target* changes this floor argument, but does not repair the finite-mixture moment obstruction when the nondegenerate Heston component remains.

Exponential characteristic decay also implies a real-analytic log-density and hence a real-analytic call curve on $K > 0$. A finite mixture with positive widths is analytic there too. Equality on an entire open strike interval would therefore extend to all positive strikes and contradict Theorem 8.1. At zero common width, the finite curve is piecewise linear and cannot equal a strictly curved Bates curve on an open interval. Exact equality at finitely many points is not excluded by either argument.

8.5 An exact continuous SANOS-type representation of Bates

The reports' exact extension replaces a finite atomic mixing measure by a generally continuous measure on *anchors and variances*. For (8.1), define the future skeleton variables

$$\begin{aligned} I_{t,T} &= \int_t^T V_u \,\mathrm{d}u, \qquad U_{t,T} = \int_t^T \sqrt{V_u}\,\mathrm{d}B_u = \frac{V_T - V_t - \kappa\theta(T-t) + \kappa I_{t,T}}{\xi}, \\ A_{t,T} &= S_t \exp\left\{\rho U_{t,T} - \tfrac{1}{2}\rho^2 I_{t,T} + J_T - J_t - \int_t^T k_J(u)\,\mathrm{d}u\right\}, \\ v_{t,T} &= (1-\rho^2) I_{t,T}. \end{aligned} \tag{8.10}$$

Theorem 8.3 (Exact conditional Black mixing)**.** *Under the reference law,*

$$\boxed{C_t^{\mathrm{B}}(T,K) = \mathbb{E}_t[\mathcal{B}(A_{t,T}, K, v_{t,T})] = \int \mathcal{B}(a,K,v)\,\mu_{t,T}(\,\mathrm{d}a,\,\mathrm{d}v),} \tag{8.11}$$

where $\mu_{t,T} = \mathrm{Law}(A_{t,T}, v_{t,T} \mid \mathcal{F}_t)$ *has unit mass and first anchor moment* $\int a\,\mu_{t,T} = S_t$.

Proof. Condition further on the future variance path and jump skeleton. The remaining integral $\sqrt{1-\rho^2}\int_t^T \sqrt{V_u}\,\mathrm{d}W_u$ is conditionally centred Gaussian with variance $v_{t,T}$. Thus the conditional terminal stock is lognormal with forward $A_{t,T}$ and total variance $v_{t,T}$. Conditional pricing followed by the tower property proves the formula. Applying the same conditioning to S_T and using its true-martingale property proves the mean constraint. □

This retains the defining positive-kernel structure of SANOS, but not the finite common-width implementation: both the anchors and widths have a mixing distribution. Structured component-dependent widths already belong to the broader SANOS framework [4]; the further enlargement here is continuous mixing. A finite-dimensional Bates model can generate this measure, even though the measure itself is not a finite vector of production SANOS parameters. Positivity and the anchor mean imply strike-wise static admissibility; cross-expiry and dynamic consistency here follow from the common oracle law, not merely from those two moment constraints.

There is also exact representation freedom. For $0 < \lambda \le 1$, let Z_1 be an auxiliary independent standard normal and set

$$A_\lambda = A_{t,T}\exp\{-\tfrac{1}{2}(1-\lambda)v_{t,T} + \sqrt{(1-\lambda)v_{t,T}}Z_1\}, \qquad v_\lambda = \lambda v_{t,T}. \tag{8.12}$$

Splitting a Gaussian into two independent parts gives

$$C_t^{\mathrm{B}}(T,K) = \int \mathcal{B}(a,K,v)\,\mu_{t,T}^{\lambda}(\,\mathrm{d}a,\,\mathrm{d}v), \qquad \mu_{t,T}^{\lambda} = \mathrm{Law}(A_\lambda, v_\lambda \mid \mathcal{F}_t). \tag{8.13}$$

For any observation dates t_n and live expiries T_m, the chronological parameter sequence is explicitly

$$\Theta_{n,m} = \mu_{t_n,T_m}^{\lambda_n}, \qquad \lambda_n = \tfrac{1}{2} + \tfrac{1}{4}\sin(t_n/t_{\mathrm{yr}}),$$

where t_{yr} denotes one year; constant $\lambda_n = 1/2$ is simpler and equally valid. These are *measure-valued formulas*, not already computed particle arrays. The component smoothing standard deviation is $\sqrt{v}$, not a second freely chosen common scalar η.

Changing λ changes the representation, not the price. Hence, wherever generator derivatives exist, their complete sum still has zero drift; all representation-change terms cancel. The conditional law and its full evolution must be changed together. The auxiliary Z_1 is integrated out and is not newly revealed market information. Finite quadrature, particle sampling or neural compression of the mixing law introduces approximation error and does not inherit an exact identity merely by using the same notation.

The reference has no scheduled jump at a deterministic expiry. As $t \uparrow T$, the conditional terminal law collapses to δ_{S_T}, integrated variance vanishes, and (8.11) converges to intrinsic value. It supplies a front-expiry benchmark without a fixed cash-node pinning requirement. The target is smooth at positive remaining maturity, whereas its terminal payoff is necessarily kinked. The conditioning argument relies on the independence in (8.1); arbitrary state-dependent CRC resets may invalidate it and require a different conditional kernel.

8.6 Fixed-oracle replication versus a freely chosen SANOS market

If the measure, filtration and underlying stock of the oracle are retained, a true-martingale call price with terminal payoff $(S_T - K)^+$ must equal $C_t^{\mathrm{B}}(T,K)$. Consequently, fitting only the Bates nodes and requiring a complete finite-Black-mixture ordinary-call market *under that same law through settlement* cannot evade Theorem 8.1. This does not prohibit exact finite-node fitting, approximate full fitting, or a different SANOS market with its own coherent dynamics.

There is a useful local control experiment. Fix anchors, a positive common width w_0, and an invertible matrix

$$M_0 = \begin{pmatrix} 1 & \cdots & 1 \\ a_1 & \cdots & a_N \\ \mathcal{B}(a_1,K_1,w_0) & \cdots & \mathcal{B}(a_N,K_1,w_0) \\ \vdots & & \vdots \\ \mathcal{B}(a_1,K_n,w_0) & \cdots & \mathcal{B}(a_N,K_n,w_0) \end{pmatrix}, \qquad N = n+2.$$

For one expiry set

$$f_t = (1, S_t, C_t^{\mathrm{B}}(T,K_1), \ldots, C_t^{\mathrm{B}}(T,K_n))^\top, \qquad \pi_t = M_0^{-1} f_t. \tag{8.14}$$

Every query $H_t(K) = b_0(K)^\top M_0^{-1} f_t$, with $b_{0,j}(K) = \mathcal{B}(a_j,K,w_0)$, is a deterministic linear combination of martingales. Where $\pi_t \geq 0$, it is a smooth positive mixture fitting the protected quotes exactly and with zero generator residual everywhere in strike. This is an explicit local parameter process, not a numerical search. Its limitations are also explicit: positivity can fail, jumps can exit the positive domain by overshoot, and at T its payoff is generally a portfolio of protected payoffs rather than $(S_T - K)^+$. Stopping a continuous process at the positivity boundary does not repair the ordinary-call terminal condition.

For a genuinely state-dependent fit, time-varying smoothing contributes more terms. With fixed anchors,

$$d\pi_j = \mu_j\,\mathrm{d}t + \psi_j \cdot \mathrm{d}W, \qquad dw = a_w\,\mathrm{d}t + b_w \cdot \mathrm{d}W,$$

the continuous price drift is

$$R_C = \sum_j \mathcal{B}_j\mu_j + a_w \sum_j \pi_j \mathcal{B}_{j,w} + \tfrac{1}{2}\|b_w\|^2 \sum_j \pi_j \mathcal{B}_{j,ww} + \sum_j \mathcal{B}_{j,w}\psi_j \cdot b_w. \tag{8.15}$$

If $w = \eta^2$ and $d\eta = a_\eta\,\mathrm{d}t + b_\eta \cdot \mathrm{d}W$, then $a_w = 2\eta a_\eta + \|b_\eta\|^2$ and $b_w = 2\eta b_\eta$. Moving anchors add their first, second and cross derivatives. With compensated jumps, add the full remainder in (8.2); also $\Delta w = 2\eta_- \Delta\eta + (\Delta\eta)^2$ and the drift of w contains the predictable quadratic-jump term. A nonsmooth calibration active-set change is not covered by classical differentiable Itô formulas without further analysis.

More generally, after fixing all coefficients except the drift a_U of an additional smoothing state U, the correction problem has the form

$$D_U H(T,K)\, a_U = -R_0(T,K) \quad \text{for every protected and off-grid contract.} \tag{8.16}$$

Instantaneous solvability requires the residual function to belong to the span of the available smoothing derivatives. One scalar drift cannot generally cancel an arbitrary function of strike and maturity. Even solvability leaves positivity, well-posedness, integrability and settlement to prove. If node agreement is an identity $H(t,z,u;T_m,K_i) = c_i^m(t,z)$ throughout the state domain, all U and mixed z,U derivatives vanish there, and the generator at that node is exactly the original grid-price generator. A pointwise numerical fit at isolated states is weaker than this functional identity.

8.7 Precise terminal rigidity, and the sufficient relaxation

The second report strengthens the discussion of the front boundary with two theorems that do not depend on Bates.

Proposition 8.4 (A positive remaining-variance floor prevents continuous settlement)**.** *Suppose $S_t \to S_T > 0$ and a positive mixture (8.5), with moving weights and anchors allowed, has mean S_t. If every active component has $w_j(t,T) \geq \varepsilon > 0$ near T, it cannot converge to $(S_T - K)^+$ for every $K > 0$.*

Proof. Monotonicity in variance and convexity in the forward argument give

$$H_t(T,K) \geq \sum_j \pi_j(t)\mathcal{B}(a_j(t),K,\varepsilon) \geq \mathcal{B}(S_t,K,\varepsilon).$$

For positive spot, strike and variance the latter strictly exceeds intrinsic value. Take $t \uparrow T$ to obtain a contradiction. □

The qualification about continuous settlement is essential. A model that keeps $S_t = s$ until T and then reveals $S_T = sZ$, for an independent mean-one lognormal Z, is a valid martingale model with a scheduled terminal surprise. Its pre-expiry Black variance need not vanish. This exception is absent in the continuous product construction and at deterministic expiries of the finite-intensity Bates reference.

Proposition 8.5 (Fixed cash anchors impose pinning and exclude a closed smooth market)**.** *For distinct fixed positive cash anchors $a_1,\dots,a_N$, suppose a common-width positive mixture converges to ordinary all-strike settlement and $S_t \to S_T > 0$. Then $w(t,T) \to 0$, S_T is one of the anchors, and the weights converge to its unit mass. If the same finite cash anchors and mean constraint represent a closed true-martingale option market throughout $[0,T]$, then every active positive-anchor component must in fact have zero variance at every pre-expiry time.*

Proof. A subsequence with w bounded away from zero contradicts Proposition 8.4. The weight simplex is compact, so any limiting subsequence gives

$$(S_T - K)^+ = \sum_j \pi_j^*(a_j - K)^+ \quad (K > 0).$$

Uniqueness of the call-implied probability law implies $\sum_j \pi_j^* \delta_{a_j} = \delta_{S_T}$, proving pinning and uniqueness of the limit. For the stronger closed-market assertion, the mean constraint bounds

S_T above by $a_{\max}$. Any call with $K > a_{\max}$ therefore has zero terminal payoff and, by true martingality, zero earlier price. An active Black component with positive variance gives a strictly positive price there, a contradiction. □

These restrictions concern fixed *cash* anchors and remaining total variance. They do not prohibit a fixed positive volatility rate with $w(t,T) = \int_t^T \sigma^2(u)\,\mathrm{d}u$. Nor do they prohibit a fixed finite number of latent anchors. The construction already proved in Section 4 makes exactly the sufficient changes:

$$a_j(t) = x_j Y_t, \qquad w(t,T) = \int_t^T \sigma_Y^2(u)\,\mathrm{d}u, \qquad \pi_j(t,T) = \mathbb{Q}(X_T = x_j \mid \mathcal{F}_t^X).$$

There is one common multiplicative movement of cash anchors, a vanishing variance clock and coherent conditional weights. The observed spot need not be pinned to any fixed cash grid. Merely sending smoothing to zero on frozen cash anchors cannot supply that conclusion: at $S_T = 102$ the barycentric law $0.6\delta_{100} + 0.4\delta_{105}$ gives a call at $K = 101$ worth 1.6, whereas intrinsic value is 1. This is the same payoff distinction underlying linear interpolation in Section 3.

8.8 A dynamic completion of a directly calibrated SANOS surface

The exact mixing law (8.11) is useful as an oracle; retaining it is unnecessary when fitting a new market model to data. At fixed anchors and kernel variances, weights can be fitted by linear equality constraints, bid–ask inequalities or a convex loss. The remaining task is to supply one coherent transition law rather than independently refit unrelated snapshots.

Theorem 8.6 (Completion of calibrated marginals, including a maturity interpolation)**.** *Let q^m, $m = 0, \dots, M$, be probability rows of common mean S_0 on a common positive finite support x, with $q^0 = \delta_{S_0}$ (include S_0 as an anchor). Let V_Y be a deterministic nondecreasing absolutely continuous variance clock starting at zero. Suppose*

$$q^{m-1} P_m = q^m, \qquad P_m \mathbf{1} = \mathbf{1}, \qquad P_m x = x, \qquad P_m \geq 0. \tag{8.17}$$

Then the initial curves $C_0(T_m, K) = \sum_j q_j^m \mathcal{B}(x_j, K, V_Y(T_m))$ admit a coherent finite-SANOS completion at the model expiries. In addition, for any continuous nondecreasing $\alpha_m : [T_{m-1}, T_m] \to [0,1]$ with endpoint values $0, 1$, a finite-state jump completion matches

$$C_0(T,K) = \sum_j [(1 - \alpha_m(T)) q_j^{m-1} + \alpha_m(T) q_j^m] \mathcal{B}(x_j, K, V_Y(T)) \tag{8.18}$$

throughout that interval, and retains a finite mixture for every later conditional expiry price.

Proof. The martingale chain with transitions P_m, revealed at the model dates or progressively as in Section 9.2, gives the first assertion by the product theorem. For the second, independently schedule one transition time in each interval with distribution function α_m. Keep the old latent state until that time; then choose the next state from its row of P_m, independently of the transition time. The conditional mean of the new state is the old state, so the bounded process X is a martingale. The information set includes whether that interval's transition has occurred, including a possible self-transition. Its marginal at deterministic T in the interval is $(1 - \alpha_m(T)) q^{m-1} + \alpha_m(T) q^m$. Multiply by independent Black Y. This proves the initial interpolation and, because every X_T has finite support, finite conditional pricing at all later dates. □

On a common finite support, the transport constraints are feasible exactly for successive marginals in convex order. Different finite supports can be embedded in their union by zero-padding. The theorem starts from a compatible calibration; it does not claim that every

continuous observed surface fits a prescribed finite family. The dates' marginals determine neither the coupling nor the transition-revelation rule. Those degrees of freedom can be estimated from dynamics while keeping the pricing construction intact. Arbitrarily resetting them to fit a new surface still needs the consistency conditions of Section 12.

8.9 An alternative continuous completion with explicit information loadings

The second report supplies another continuous front construction, complementary to Brownian threshold revelation. Let a hidden terminal variable A take positive values x_j with probabilities p_j, and choose distinct signal vectors $g_j \in \mathbb{R}^r$. Independently of the smoothing factor, observe

$$I_u = g(A)u + B_u^{(r)},$$

where $B^{(r)}$ is Brownian and independent of A. Use a differentiable increasing clock $h(t)$ with $h(0) = 0$ and $h(t) \uparrow \infty$ as $t \uparrow T$, for example $h(t) = \gamma^2 t/(T-t)$. Bayes' formula gives

$$\pi_j(t) = \frac{p_j \exp\{g_j \cdot I_{h(t)} - \frac{1}{2}\|g_j\|^2 h(t)\}}{\sum_k p_k \exp\{g_k \cdot I_{h(t)} - \frac{1}{2}\|g_k\|^2 h(t)\}}, \qquad X_t = \sum_j x_j \pi_j(t). \tag{8.19}$$

The pricing filtration contains the observations, not the simulated hidden label A. Revealing A prematurely would change prices. The posterior processes are bounded conditional-probability martingales. Since $I_u/u \to g(A)$ and the signals are distinct, they converge to the indicator of the realised terminal state. In the observation filtration their diffusion is

$$d\pi_j = \sqrt{h'(t)}\pi_j(g_j - \bar{g}_t) \cdot \mathrm{d}W_t^I, \qquad \bar{g}_t = \sum_j \pi_j g_j, \tag{8.20}$$

where W^I is the innovation Brownian motion. Equation (8.20) follows by differentiating the posterior with $dI_{h(t)} = \bar{g}_t h'(t)\,\mathrm{d}t + \sqrt{h'(t)}\,\mathrm{d}W_t^I$. For scalar signals $g_j = x_j$, this gives $dX_t = \sqrt{h'(t)}\,\mathrm{Var}_{\pi_t}(x)\,\mathrm{d}W_t^I$. This is an explicit finite-signal construction within the information-based pricing approach [31].

For a finite expiry tower, use row i of P_m as the prior on interval (T_{m-1}, T_m) when the previous revealed state is x_i, and propagate later conditional rows by

$$q^\ell(t) = \pi^m(t) P_{m+1} \cdots P_\ell, \qquad \ell \geq m.$$

The resulting X is continuous across revelation dates and bounded. Its independent Black product gives

$$C_t(T_\ell, K) = \sum_j q_j^\ell(t) \mathcal{B}(x_j Y_t, K, V_Y(T_\ell) - V_Y(t)).$$

The product proof applies, with the diffusion-only Wiese–Wissel projection at the model expiries. Finite common-width closure at every intermediate expiry is not asserted for this continuous construction.

This alternative adds controllable information loadings and learning clocks while leaving the initial marginal fit unchanged. A small signal dimension uses $O(Nr)$ loading coordinates and r information drivers, plus the independent Black driver. It does not realise every desired DLV covariance field. At requested pre-expiry dates one can simulate the hidden label and observation increments exactly, evaluate the posterior by log-sum-exp, and use the revealed label at the endpoint. This avoids an Euler approximation of the singular terminal-clock SDE, while retaining the correct observation filtration for pricing.

8.10 Parsimony, the SSVI comparison and the remaining computation

The reports identify a practical distinction between fitting marginals and fitting dynamics. With M expiries and N anchors, marginal weights use $O(MN)$ coordinates. Unrestricted transport matrices require $O(MN^2)$ storage. A low-dimensional martingale generator or the $O(Nr)$ information-loading specification restricts attainable dynamics but reduces complexity. At fixed candidate anchor–variance pairs, even a larger kernel dictionary has convex weight fitting; freely fitting all anchors, widths and weights is nonlinear and often weakly identified. The exact Bates mixing law generally requires growing quadrature or conditional sampling. It is a benchmark, not a requirement for an economical DYSANOS implementation. No measured speed advantage is asserted without a benchmark.

There is a precise but limited connection with symmetric SSVI. In the family studied by El Amrani–Jacquier–Martini [32], their drift restriction makes the product of the two smile parameters invariant. Under their static-admissibility and terminal conditions, a nonzero invariant cannot persist through expiry; the surviving consistent case is flat. Our positive-variance-floor and fixed-cash-anchor propositions have a similar logical structure: specified restrictions conflict with the settlement boundary. They do not extend to all finite SANOS dynamics, as the product construction explicitly demonstrates. Nor is changing from variance to price coordinates by itself a solution: at an admissible interior call price, implied volatility is an invertible coordinate change. The substantive issues are the allowed stochastic family, its conditional-pricing consistency and its terminal boundary.

The additional Bates programme is therefore specific:

1. Reconcile (8.3) on every protected node and live non-front slab, with identical prices, finite-coordinate conventions, contract clock and jump compensation. Refine state derivatives and jump quadrature separately from the strike lattice.
2. Recompute selected-fit values, derivatives and R_C from *one identical decoder*. Record active-set changes, nonsmooth states and exclusions. Use (8.4) to cross-check call and density-coordinate residuals.
3. Solve exact quote, mass and mean constraints, recording the positive smoothing interval, conditioning and cross-expiry feasibility. Distinguish an equality solve from a small least-squares error.
4. Use (8.14) as a zero-drift local control, reporting both positivity-domain exits and its terminal-payoff failure. Use the product or posterior model as the separate full-maturity control.
5. Approximate (8.11) and (8.13) with increasing quadrature or sample resolution and an independent Fourier-pricer comparison. Check prices, moment constraints, representation invariance and near-expiry behaviour; a finite sweep cannot prove universal existence or nonexistence.
6. Export chronological finite-approximation arrays alongside their conditional-law formulas, parameter conventions, generator residuals, errors and settlement values. For an approximate CRC reset, export the full pre-/post-reset discrepancy as a separate object.

The tail theorem, local feasibility theorem and dynamic constructions answer different questions. Together they support a direct SANOS research programme without requiring exact finite-family Bates replication, while retaining Bates as a stringent independent oracle.

9 Front expiry and an explicit continuous construction

9.1 The exact finite-book boundary without spot pinning

Let an integrable martingale S take values in $[K_0, K_{N+1}]$. For $s \in [K_k, K_{k+1}]$, define

$$h_k(s) = \frac{K_{k+1} - s}{K_{k+1} - K_k}, \qquad h_{k+1}(s) = \frac{s - K_k}{K_{k+1} - K_k}, \quad h_i(s) = 0 \text{ otherwise.} \tag{9.1}$$

At a node the definition is its point mass. The barycentric law has mass one, mean s, and

$$\sum_i h_i(s)(K_i - K_n)^+ = (s - K_n)^+ \quad \text{for every grid strike } K_n. \tag{9.2}$$

The identity holds because each grid-strike payoff is affine on each grid cell.

Proposition 9.1 (Two-survivor terminal codebook)**.** *The processes $q_i(t) = \mathbb{E}[h_i(S_T) \mid \mathcal{F}_t]$ are bounded martingales, have common mean S_t, and represent the true prices of all grid calls. They converge to a two-survivor barycentric codebook at expiry, without requiring S_T to be a grid node.*

Proof. Conditional expectation preserves nonnegativity, mass, the barycentric mean and (9.2). At time T the conditional expectation is the terminal vector itself. □

This is a finite-instrument boundary construction, not an all-strike terminal distribution. Off-grid calls require more information, as the example in Section 3 shows. Nor does the proposition make an arbitrary separately specified back-month SDE compatible with this front. It supplies a correct terminal target and an existence construction once a common spot law is chosen.

9.2 Brownian revelation of an entire finite density tower

Here is an explicit continuous martingale that couples the front to the back expiries. Take a finite Markov martingale chain $(Z_1, \dots, Z_M)$ on the positive anchors x_i, with prescribed convex-ordered marginals and transition matrices P_m satisfying $P_m x = x$. Let $Z_0 = S_0$ be deterministic, permitting a separate initial state. Choose a Brownian motion B under $\mathbb{Q}$. On each interval $[T_{m-1}, T_m]$, set

$$U_m = \Phi\left(\frac{B_{T_m} - B_{T_{m-1}}}{\sqrt{\Delta T_m}}\right),$$

and select Z_m from the row of P_m associated with Z_{m-1} by inverse cumulative probabilities. The U_m are independent uniforms, so this realises the desired Markov chain.

During the interval, condition on the known previous state. Let p_j be its transition probabilities, $F_j = \sum_{i \le j} p_i$, and

$$b_j = \sqrt{\Delta T_m}\, \Phi^{-1}(F_j), \quad b_{-1} = -\infty, \quad b_L = +\infty, \quad z_t = B_t - B_{T_{m-1}}, \quad v_t = T_m - t.$$

The conditional probabilities of the next node are explicitly

$$r_j(t) = \Phi\left(\frac{b_j - z_t}{\sqrt{v_t}}\right) - \Phi\left(\frac{b_{j-1} - z_t}{\sqrt{v_t}}\right). \tag{9.3}$$

Each solves the backward heat equation; hence $dr_j = (\partial_z r_j) dB_t$. Define

$$X_t = \sum_j x_j r_j(t), \qquad q^\ell(t) = r(t) P_{m+1} \cdots P_\ell, \quad \ell \ge m. \tag{9.4}$$

Empty products are identities. Equivalently $X_t = \mathbb{E}[Z_M \mid \mathcal{F}_t^B]$. As $t \uparrow T_m$, $r(t)$ converges almost surely to the selected node; the Brownian endpoint hits a threshold with probability zero. The

next interval begins with conditional mean Z_m, so X is continuous across the reset. It is bounded and a true martingale. All future densities in (9.4) are conditional laws of this same process.

Multiplying this X by the independent Y gives a continuous spot process with the finite SANOS formula at every model expiry. The short-end is therefore settled by construction, while the latent front collapses to one atom and the observed cash spot $X_{T_m}Y_{T_m}$ remains off a prescribed fixed cash lattice. There is no need to invent a separate exploding vol-of-density heuristic to make settlement occur. The conditional probability derivatives can nevertheless become large close to expiry, so a numerical scheme should use (9.3) directly rather than Euler-discretise its singular-looking coefficients.

At maturities *between* the prescribed dates, this continuous X_T generally has a continuous law. Formula (4.4) still defines an arbitrage-free surface, but its mixture is then an integral rather than a fixed finite sum. The chain example instead retains a finite sum for all maturities at the cost of jumps. These are complementary constructions; combining their advantages without qualification would overstate the theorem.

9.3 Limits of the front resolution

The product theorem resolves terminal consistency for the constructed law. It does not prove that every empirical front density, arbitrary back-month factor field and chosen smoothing level can be combined. That is a constrained calibration and existence problem. Likewise, a latent finite-support martingale must be absorbing if it reaches an extreme support point: a random variable bounded below by x_0 with conditional mean x_0 is constant. The Black factor prevents the *observed* spot from becoming constant, but the latent smile component can still degenerate. Tail or support design remains an economic modelling issue.

10 Regridding, rolling contracts and price-preserving changes

A coordinate change and replacement of a conditional law must be kept distinct. If an inherited cash support is sx_j, and a new reference spot is sx_i, then the relative coordinates $x'_j = x_j/x_i$ satisfy

$$(sx_i)x'_j = sx_j.$$

All inherited cash payoffs and their prices are unchanged pathwise. No interpolation theorem or grid-overlap condition is needed for this identity.

For genuine support replacement with proposed nodes z_j, the exact conditions for a protected book with payoff functions f_a are

$$\widetilde{q}_j \geq 0, \qquad \sum_j \widetilde{q}_j = 1, \qquad \sum_j \widetilde{q}_j z_j = S, \qquad \sum_j \widetilde{q}_j f_a(z_j) = C_a \quad \text{for every protected } a. \tag{10.1}$$

For a smoothed mixture replace $f_a(z_j)$ by the corresponding Black kernel, and include maturity couplings. Feasibility is a convex-hull question. If every proposed cash node lies strictly on one side of the old support, the old mean cannot be retained. Nodes outside the old range on *both* sides can bracket its mean, but that is not sufficient to preserve calls. Masses $(1/4, 1/2, 1/4)$ at $(80, 100, 120)$ have mean 100 and $C(100) = 5$; equal masses at $(70, 130)$ retain the mean and give $C(100) = 15$.

When the regridding operation samples an inherited convex call curve at new strike knots and linearly interpolates those samples, an old protected strike is preserved if retained, or if its containing new interval has no genuine inherited kink. Crossing a kink gives a strict chord increase in the interior. Retaining every strike price of an atomic law requires retaining the full law, equivalently all genuine call-curve kinks, not merely its mean. Adaptive support design is therefore naturally tied to an explicit protected book.

Rolling time to maturity does not change the identity of contracts already held. At a roll, expired contracts settle, surviving (T, K) remain the same, and newly introduced back-expiry contracts require an admissible extension. Comparing an old four-week call with a newly issued four-week call is not a martingale test on one asset. A newly added expiry can be priced from an extended coherent martingale law; an independent statistical forecast of its DLV curve is not by itself such an extension.

11 Physical dynamics, admissible risk premia and factor dimension

11.1 Constructing P around an admissible Q model

The measure Q prices contingent claims; the physical measure P describes the distribution of market scenarios used for estimation, forecasting and hedging performance. A useful historical simulator needs both. A good fit to a sequence of daily option surfaces does not identify either their transition law or the market price of risk uniquely.

Let a continuous state representation of the constructed model be

$$dZ_t = b^Q(t, Z_t)dt + \Sigma(t, Z_t)dW_t^Q, \qquad C_t^a = H_a(t, Z_t),$$

with $\mathcal{L}^Q H_a = 0$ for every protected fixed contract a. Adopt the sign convention

$$W_t^Q = W_t^P + \int_0^t \lambda_s ds, \qquad b^P = b^Q + \Sigma\lambda. \tag{11.1}$$

On a finite horizon, assume the stochastic exponential

$$D_t = \left.\frac{dQ}{dP}\right|_{\mathcal{F}_t} = \mathcal{E}\left(-\int_0^{\cdot} \lambda_s \cdot dW_s^P\right)_t \tag{11.2}$$

is a strictly positive true martingale. Bounded λ, or an appropriate Novikov condition, is sufficient in this Brownian setting. Conversely, starting directly under Q, one may define P by the true exponential $\mathcal{E}(\int \lambda\, dW^Q)$ and obtain the same convention.

With $A_a = DH_a\Sigma$, the physical drift of a fixed-contract price is

$$\mu_a^P = A_a\lambda. \tag{11.3}$$

Thus admissible risk premia may be estimated from historical transitions without destroying the constructed martingale measure. Static admissibility is a path property and remains valid under an equivalent measure. Independence of the factors is required under the pricing construction Q; it need not persist under P after a state-dependent likelihood tilt.

For the stochastic smoothing model (5.4)–(5.5), two components of the market price of risk give, in the same convention,

$$\frac{dY}{Y} = \sigma\lambda_1 dt + \sigma dB^{1,P}, \qquad b_V^P = b_V^Q + a(\rho\lambda_1 + \sqrt{1-\rho^2}\lambda_2).$$

The second Brownian direction can carry an additional volatility-state risk premium. This is a genuine enlargement of the model's diffusion span; it does not make an arbitrary drift vector in the full DLV state attainable. A stochastic volatility-of-volatility state adds its own generator and, where independently driven, another risk-premium direction.

11.2 Tradable options remove rows; factor reduction removes directions

For a traded set $\mathcal{J}$, stack the volatility rows into $A_{\mathcal{J}}$ and the physical drifts into $\mu^P_{\mathcal{J}}$. An equivalent Brownian drift change requires

$$\mu^P_{\mathcal{J}} \in \operatorname{Ran} A_{\mathcal{J}}, \qquad \lambda = A^+_{\mathcal{J}}\mu^P_{\mathcal{J}} + v, \quad v \in \ker A_{\mathcal{J}}. \tag{11.4}$$

The smallest-norm candidate and its unexplained component are

$$\lambda_{\min} = A^+_{\mathcal{J}}\mu^P_{\mathcal{J}}, \qquad r_{\mathcal{J}} = (I - A_{\mathcal{J}}A^+_{\mathcal{J}})\mu^P_{\mathcal{J}}. \tag{11.5}$$

A nonzero residual rules out such a measure change for those stipulated local coefficients. A zero residual is only the algebraic step; the exponential still has to define an equivalent probability measure.

Modelling operation	Linear-algebra effect	Consequence
Remove traded options	Delete rows of A and entries of μ	Fewer drift equations; a reduced market may admit more risk premia.
Reduce Brownian drivers	Restrict the column space of the diffusion	Fewer available drift-change directions; compatibility usually becomes harder.
Add an independent Black factor	Add a particular sensitivity column to the price volatility	Adds one economic risk direction; does not generally span missing latent DLV drift directions.
Compress the state decoder	Restrict reachable price surfaces and their tangent directions	Must preserve the generator constraint, not only reconstruction accuracy.

In particular, reducing Brownian drivers is *not* the operation that removes tradable options. Hundreds of prices can be driven consistently by a single Brownian motion, as in a Black model. What they cannot have is hundreds of independently chosen physical drifts with the same one-dimensional loading span.

At the state level the stronger test is

$$b^P_{\text{target}} - b^Q \in \operatorname{Ran} \Sigma. \tag{11.6}$$

A drift mismatch invisible to a reduced traded book can fail (11.6) while passing its price-level analogue. Conversely, starting with $b^P = b^Q + \Sigma\lambda$ avoids that inconsistency by construction. This is why joint constrained estimation is preferable to estimating an unrestricted statistical drift and attempting to remove it afterwards. The admissible directions form a linear span in the unconstrained Brownian setting, not a cone.

If the entire smooth surface is declared tradable, (11.3) is a functional restriction over all (T, K). A smaller protected book weakens the theorem about the market. It does not justify claiming no dynamic arbitrage for the unprotected continuum. In the product construction all its quoted calls already share Q, so adding them to the traded set preserves the guarantee; the issue is whether a separately fitted P transition can still share that same measure.

11.3 Jumps and support restrictions

For a state with compensated jumps,

$$dZ = b^P dt + \Sigma dW^P + \int \chi(e)(\mu(dt, de) - \nu^P_t(de)dt),$$

a measure change may alter both Brownian drift and jump compensator. With $\nu^Q = r\nu^P$, equivalence requires a strictly positive density r on the relevant support and integrability of the full likelihood process. In the same truncation convention,

$$b^Q = b^P - \Sigma\lambda + \int \chi(e)(\nu^Q - \nu^P)(de). \tag{11.7}$$

The Brownian span condition alone is then incomplete. Nevertheless an equivalent measure cannot create transitions that have zero probability under the other measure. In the finite-state example, permitted physical intensities can be estimated by positive tilts of the risk-neutral transition intensities, with the complete likelihood recorded. A structural zero in the chosen chain remains a structural zero.

11.4 Low-dimensional or neural representations

For a general smooth decoder $H(t, z; T, K)$, fixing Σ requires solving

$$DH\, b^Q = -\partial_t H - \tfrac{1}{2} D^2 H : (\Sigma\Sigma^\top) \tag{11.8}$$

for every traded contract, with the jump generator added if present. A low-dimensional state must have enough tangent directions to represent the entire required right-hand side. A neural network can parameterise coefficients, likelihood tilts or an admissible conditional law, but a small training residual does not prove the functional identity. The product representation avoids solving an arbitrary off-grid correction problem because its pricing kernel satisfies the backward equation exactly.

There is no general impossibility theorem saying that a fixed finite number of SANOS anchors and a fixed nonzero smoothing coefficient preclude dynamic no-arbitrage: (4.12) is a counterexample. Conversely, this counterexample does not imply that a prescribed compressed decoder can reproduce every historical surface or every Bates law. Representation error, coefficient admissibility and dynamic arbitrage are separate questions.

12 A CRC-inspired consistent calibration programme

12.1 The useful lesson from generalised Bates CRC

In Gambara–Teichmann's construction [15], the current extended affine model first evolves under its admissible pricing dynamics. Structural parameters may then be changed, while the extension is recalibrated so that the *already evolved* surface is represented by the new parameters. The extension is a codebook correction, not an independent source of arbitrary price changes. Their Bates implementation uses an inverse calibration step for the jump extension; an approximate inverse has to be distinguished from an exact consistency theorem.

The analogue here is to separate a conditional price law from its coordinates. Latent transition or revelation parameters govern X; a clock or independent kernel governs Y; mixture probabilities and DLV coordinates encode the resulting surface. Historical observations determine which admissible dynamics are useful. They do not authorise a price discontinuity every time a fitting algorithm changes parameters.

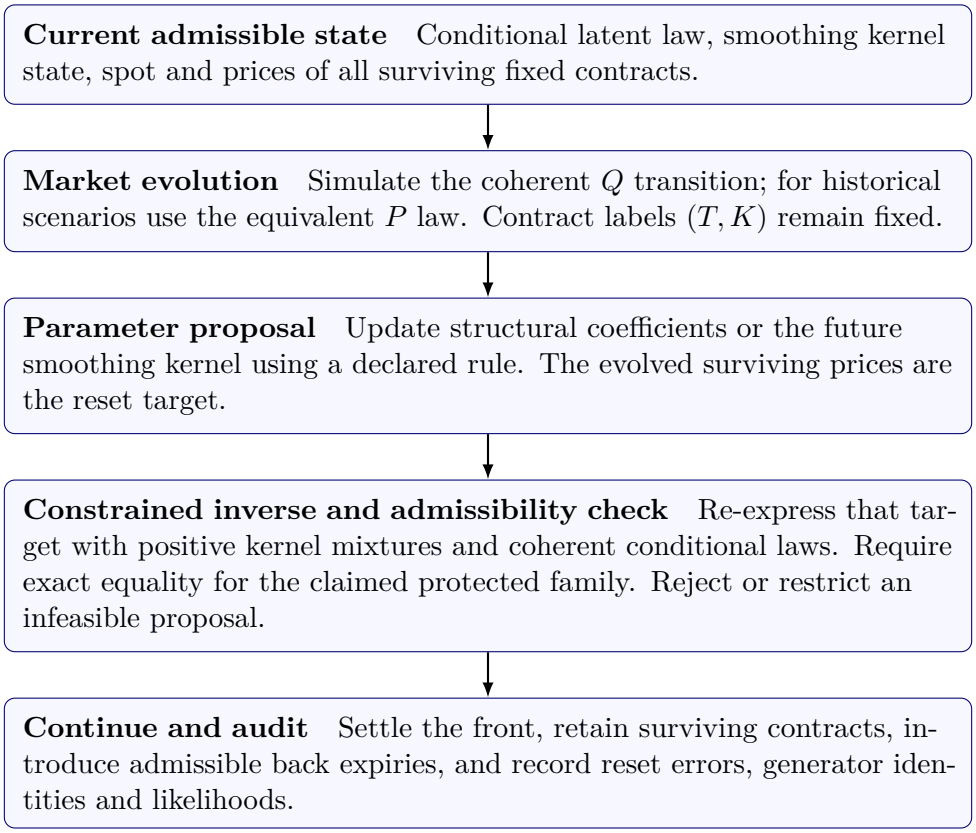


Figure 1: Consistent recalibration in the SANOS/DYSANOS coordinates. The price move is produced by the stochastic transition; the subsequent coordinate reset preserves the evolved prices.

12.2 A stitching theorem for price-preserving resets

Proposition 12.1 (Finite consistent recalibration)**.** *Consider finitely many ordered reset times before T_*. On each interval, choose an admissible conditional product model, or another common martingale model, for spot and a protected family of surviving contracts. Assume measurable conditional transition kernels, integrability sufficient for true martingality, and correct contractual settlement. At each reset, let the new model start from the old spot and the old values of every protected surviving contract, pathwise. Then the concatenated prices are martingales under the law obtained by concatenating these Q kernels. If the protected family contains all strikes and relevant expiries, the full stated surface retains dynamic no-arbitrage.*

Proof. For two times within an interval, the conditional martingale identity is part of the chosen model. At a reset, the price before and after coordinate replacement is equal. Applying conditional expectation across that reset therefore leaves the pre-reset value unchanged. Induction over the finite number of reset times proves the tower property across arbitrary intervals. Settlement identifies the final cashflow. The same law is used for the whole protected family. □

This is a sufficient stitching principle, not a theorem that the required inverse always exists. Preserving finitely many quotes preserves only that finite protected family at the reset. Full continuum preservation requires equality of the conditional call laws, equivalently their strike distributions at each surviving expiry, and can be much more restrictive. If a reset is intended to be an actual newly revealed market shock rather than a coordinate change, its price jump must instead satisfy the corresponding conditional martingale restriction; it cannot be declared price-preserving.

12.3 An implementable order of estimation

A constrained historical estimation problem can be written schematically as

$$\min_{\vartheta,\lambda,z_{0:n}} \sum_{k,a} \mathcal{E}_{\text{quote}}\Big(H_a(t_k, z_k; \vartheta), c^{\text{obs}}_{k,a}\Big) - \sum_k \log p^{P,\vartheta,\lambda}_{\Delta t_k}(z_{k+1} \mid z_k) + \mathcal{R}(\vartheta, \lambda, z_{0:n}), \tag{12.1}$$

subject to the exact structural constraints described below. The quote error can vanish inside bid–ask intervals. The transition density may be replaced by a justified likelihood approximation or another proper statistical score. $\mathcal{R}$ controls complexity, roughness and risk-premium size; it is not a replacement for martingale constraints.

The practical sequence is:

1. **Fix the market and coordinates.** Declare the protected contracts, cash strikes, calendar expiries, observation times, tail treatment and settlement rule. Separate latent anchors from observed strikes. Decide whether the all-maturity chain model or the continuous model-expiry construction is used.
2. **Fit static states jointly with smoothing.** For proposed anchors and clock, solve the positive kernel calibration with mass, mean and latent convex-order constraints. With fixed kernels these are linear constraints. Retain bid–ask feasibility information and distinguish exact equality from approximation.
3. **Fit admissible risk-neutral transitions.** Choose martingale couplings, chain generators satisfying $Ax = 0$, or Brownian revelation parameters. Require an actual conditional density tower. Derive the induced DLV drift rather than estimate it as a free parameter. If starting from arbitrary DLV loadings, verify existence, positivity and front closure separately.
4. **Fit physical risk premia.** Estimate λ and, if relevant, positive jump-intensity tilts. Set the physical drift through (11.1) or (11.7). Check span, conditioning and likelihood integrability. Record P- and Q-dynamics in the same state coordinates.
5. **Refine the historical fit jointly.** Adjust latent states, the admissible transition family and nonidentified smoothing choices using (12.1). Repeated daily fitting with unconstrained parameter jumps is not a dynamic model. If the data cannot be explained within the current family, enlarge that family or report the misspecification.
6. **Specify the reset rule before simulation.** Simulate market evolution first; preserve its resulting prices during any parameter re-expression. Introduce new expiries through admissible extensions. If only a finite book is protected, state that restriction in the reported guarantee.
7. **Evaluate independently.** Test quote fit, forecast and hedging performance on held-out dates and regimes. Audit exact identities before using OCE or Monte Carlo mean tests.

In observed data, today's surface is exogenous information to a filtering and estimation procedure. In a simulator, today's surface must be generated by its stochastic law. These roles should not be confused. A filter can update beliefs after an observation; a simulated price cannot be manually moved to a desired target and still inherit a martingale proof merely from static recalibration.

12.4 What is fixed and what is recalibrated

Quantity	During a simulated interval	At a coordinate reset
Surviving (T, K)	Fixed contractual identity	Retained until settlement
Spot and call prices	Evolve under the selected P or Q law	Preserved for the protected family
Latent probabilities	Conditional-law evolution, not arbitrary fitting	May change if the same protected prices and a coherent continuation can be represented
DLV coordinates	Derived from the relevant latent or observed call book	Recomputed from that same book
Smoothing kernel	Additive Black clock, or an enlarged stochastic-volatility state with its full generator	Future kernel may change only with an admissible compensating recalibration
Risk-neutral coefficients	Prescribed admissible transition rule	May be updated within the admissible family
Physical risk premia	Adapted coefficients of an equivalent measure	May be re-estimated; likelihood and support conditions still apply

This is a programme for consistent estimation, not a guarantee of statistical accuracy. In particular, no-static-arbitrage is different from "no statistical arbitrage". Under P, economically meaningful risk premia and positive expected returns can remain.

13 Verification: identities, numerical schemes and sampled markets

13.1 A hierarchy that separates causes of failure

The construction provides an analytical statement. A discretisation and an estimated generator are additional objects. Sampling error can make a correct model look inconsistent, while a weak test can miss genuine bias. A defensible audit therefore proceeds in the following order.

1. **Contract and law consistency.** Check fixed cash strikes and expiries, front settlement, tails, and the same underlying terminal law for spot and calls. Verify mass, mean, conditional-tower and price-preserving reset equalities.
2. **Deterministic drift identities.** Use one decoder, one generator, one clock and one jump convention for prices and all derivatives. Check (2.5), (3.1), (7.5) or (7.8) on all protected nodes and an independent dense off-grid set. A failure of an exact coordinate identity is not explained by coarse strike spacing.
3. **Boundary and numerical regularity.** Identify zero densities, near-singular inverse calibrations, nonsmooth active-set changes and maturity limits. Report excluded states. Confirm the claimed convergence under time-step refinement and independent derivative calculations.
4. **Conditional Monte Carlo moments.** Test the full protected book, including contracts near the tails and near expiry. Use independent path blocks and economically scaled confidence intervals. A large p-value does not prove equality, and an isolated small p-value among many tests does not by itself identify a model defect.
5. **Economic challenger tests.** Train hedging or OCE strategies independently, evaluate on held-out paths, and separate statistical risk premia from pathwise arbitrage. Check weight concentration and rare-event coverage before interpreting reweighting failures.

The all-maturity chain example permits exact transition simulation and exact Black increments, so its martingale audit need not be contaminated by an Euler scheme. The Brownian revelation example permits direct evaluation of conditional probabilities. These are useful benchmark implementations against which a more flexible learned simulator can be tested.

13.2 OCE as a diagnostic and a measure selector

Let ΔC_{k+1} denote gains of the *same* fixed contracts, including settlement when an expiry is crossed. For a predictable strategy a, write $G(a) = \sum_k a_k \cdot \Delta C_{k+1}$. Let u be increasing and concave, with $u(0) = 0$ and $u'(0) = 1$. The optimised certainty equivalent problem is

$$V_P = \sup_{a,c}\{\mathbb{E}_P[u(c + G(a))] - c\}. \tag{13.1}$$

Under the constructed Q, for strategies with integrable zero-mean gains, concavity gives $u(x) \le x$, hence $V_Q \le 0$. The zero strategy and $c = 0$ attain zero. Therefore the ideal population benchmark is $V_Q = 0$. Under P, V_P can be positive because of ordinary risk premia; this is not by itself a classical arbitrage.

Proposition 13.1 (The interior OCE first-order condition)**.** *Suppose a finite optimum (a^*, c^*) exists, $u' > 0$, differentiation under expectation is justified, and the strategy space allows both signs of all bounded predictable perturbations in the protected assets. Then*

$$D^* = u'(c^* + G(a^*)) > 0, \qquad \mathbb{E}_P[D^*] = 1, \tag{13.2}$$

and $dQ^* = D^* dP$ makes the protected discrete-time gains conditionally mean zero.

Proof. Differentiating in c yields $\mathbb{E}_P[D^*] = 1$. A perturbation that holds the jth asset over one period on an arbitrary event $A \in \mathcal{F}_{t_k}$ yields $\mathbb{E}_P[D^* \mathbf{1}_A \Delta C^j_{k+1}] = 0$. This is equivalent to the conditional martingale increment identity under Q^*, since the conditional density is strictly positive. Positivity and normalisation give an equivalent probability measure. □

This is the frictionless population mechanism behind OCE-based drift removal [27]. A neural strategy class tests only its available perturbations. Transaction costs or trading constraints change the conclusion to inequalities or a near-martingale statement. Existence of a finite optimum and the required moments are substantive assumptions; a finite training sample does not establish them. In an incomplete market the selected Q^* also need not equal the product construction's chosen Q.

For exponential utility, optimising over the cash shift gives $-\gamma^{-1} \log \mathbb{E}[e^{-\gamma G}]$. In the special one-period Gaussian case, $G = aR$, with mean μ and variance $v > 0$, maximisation over a gives $\mu^2/(2\gamma v)$. OCE is therefore not generally "mean divided by variance". Exponential integrability can fail for unbounded short-option losses; the utility and strategy class must be selected accordingly.

13.3 Why span and ESS are separate diagnostics

The span residual (11.5) asks whether an equivalent drift change is possible for the specified coefficients. Effective sample size asks whether a *particular finite sample* can estimate expectations under a chosen density ratio:

$$\mathrm{ESS} = \frac{(\sum_{n=1}^N D_n)^2}{\sum_{n=1}^N D_n^2}. \tag{13.3}$$

For constant Brownian λ and horizon T, $\mathbb{E}_P[D^2] = e^{\|\lambda\|^2 T}$, so the large-sample ESS fraction is $e^{-\|\lambda\|^2 T}$. An equivalent measure may exist while the effective sample is tiny. Conversely, no increase in sample size repairs a structural nonzero span residual for an incompatible prescribed diffusion drift.

For an independent one-step sample, strictly positive weights satisfying all empirical mean-zero constraints exist exactly when zero lies in the relative interior of the convex hull of the observed return vectors. A missing rare payoff event can make this empirical feasibility fail even when the population model is arbitrage-free. That failure concerns the finite empirical

distribution. It is not a proof that the analytical law permits arbitrage. Dynamic feasibility must use the appropriate information and conditional branches; a path cloud with a unique observed successor at every state can create artificial predictability.

When the likelihood ratio is known from the fitted model, a useful out-of-sample family of checks is

$$\mathbb{E}_P\left[D_{T_*}\, f(Z_{t_k})\Delta C^a_{k+1}\right] = 0 \tag{13.4}$$

for selected conditioning features f and every protected asset a. Weight concentration, tail-event counts, simultaneous uncertainty and solver accuracy should accompany the residuals. These checks complement the structural proof; they do not replace it.

14 Discussion and research implications

The central conclusion is constructive. A finite density market and a smooth surface need not be competing descriptions. A coherent latent martingale, an independent smoothing martingale and conditional pricing give one common market. The Wiese–Wissel condition is its exact finite-grid coordinate restriction; the continuous covariance drift is the corresponding smooth identity, with the correct clock, closure and jump terms.

Section 8 sharpens the role of an affine oracle such as generalised Bates: its nondegenerate marginal cannot be represented by a finite Black mixture, whereas a continuous anchor–variance mixture is exact. These representation results do not obstruct a different coherent finite SANOS market. An oracle is valuable for testing decoders, numerical derivatives and drift calculations against a known law. Exact representation of its full surface is not required to build an arbitrage-free SANOS-type market. Nor does a good static approximation to that oracle inherit its dynamics. The appropriate test is whether the *fitted decoder under its declared generator* has zero drift and correct settlement. The Black product model supplies an independent constructive benchmark for that test.

Common-node matching has a precise scope. Positive smoothing with frozen atomic weights changes prices. Joint calibration can restore protected quotes, and a strict finite book admits a nonzero smoothing interval under Theorem 6.2. Thereafter a coherent law determines both its grid and off-grid evolution. Preserving an externally specified full DLV trajectory, or changing a finite-mixture clock while preserving a whole continuum of quotes, remains a stronger inverse problem and may be infeasible.

The resulting historical calibration problem is constrained but not circular. Admissible Q transitions determine the allowed drift geometry; risk premia choose compatible P dynamics; cross-sectional observations and transition statistics select among the remaining freedoms. Reducing tradables, reducing state dimension and reducing Brownian drivers are distinct choices. CRC-inspired resets are useful only when their correction preserves the family for which the no-arbitrage claim is made.

The main open implementation questions are quantitative: efficient estimation of a sufficiently rich latent transition family, robust inverse calibration at large smoothing levels, long-horizon support management, and stable simulation close to settlement. These do not negate the proved existence examples. They determine whether the guaranteed model family is rich and economical enough for useful deep-hedging experiments.

A A compact specification for a reproducible implementation

An implementation should export the following objects separately rather than store only surface snapshots:

1. The latent anchor array x, protected cash strikes K, calendar expiries T, and the time grid. Record the distinct meaning of latent weights and observed butterflies.

2. The additive Black variance clock, latent initial probabilities and martingale transition matrices or generator. For the Brownian construction, store the transition thresholds in (9.3).
3. The kernel calibration equations, their exact or bid–ask targets, positivity and convex-order margins, and inverse conditioning diagnostics.
4. The decoder, its analytic or validated automatic derivatives, and both Q and P generators in one convention. Store the full measure-change likelihood, including reset or jump terms where applicable.
5. Every original surviving contract's chronological prices and settlement. A floating display grid can be exported as an additional view, never as a replacement for the fixed-contract audit.
6. The full set of structural residuals, conditional sample moments, independent OCE evaluation, ESS and rare-event coverage. Report all exclusions and the reasons for them.

For the finite-chain benchmark, simulate the chain using its exact time-homogeneous transition matrix or a valid nonhomogeneous-chain algorithm, and simulate Black increments exactly. Reprice by (4.10). For the revelation benchmark, simulate Brownian increments and evaluate (9.3)–(9.4) directly. These benchmarks distinguish a model error from an error introduced by a generic SDE discretisation.

A minimal verification example uses (4.12) with $r > 0$, $\sigma_Y > 0$, several fixed expiries, and a dense set of cash strikes. The exact generator residual is zero by (4.11); this is an analytic identity, not a reported numerical outcome. The finite butterfly and calendar ratios must reproduce the jump version of the Wiese–Wissel condition when evaluated from this same decoder. A diffusion-only residual would be the wrong diagnostic for this example.

B Attribution and the status of the combined results

This manuscript consolidates the supplied density-DLV, exact Black-product, Carmona–Nadtochiy/CRC, and extended calibration notes, together with the two Bates/SANOS analyses dated 23 and 24 September 2026 [28, 29]. The presentation supplies motivation and background only. The finite-density derivation is credited to Wiese's discussion note [10]; the reconciliation identifies its established Wissel predecessor. The static product interpretation is already part of the SANOS lineage and is not claimed here as a newly discovered static parametrisation.

The mathematical additions developed explicitly in this manuscript are the single coherent conditional-law formulation, its settlement and between-strike generator calculations, the concrete finite-chain and Brownian-revelation examples, and the local positive-smoothing exact-fit argument with its stated limitations. Section 8 incorporates the additional proofs and examples from those analyses; Table 1 attributes their numerical observations without claiming a fresh run or inspection of the underlying code. The historical calibration procedure is a research methodology constrained by those results; its statistical effectiveness still requires experiments. The exact numbered reference to Wissel's 2007 equation follows the supplied reconciliation note; the full algebraic relation is restated in (2.7) so the proof does not rely on an inaccessible equation label. A submission version should archive the exact 2007 source alongside that reference.

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